A representational analysis of Czech palatalization

Edoardo Cavirani Guido Vanden Wyngaerd

KU Leuven

Abstract

We discuss three patterns of palatalization in Czech, each of which is associated with certain suffixes. The data suggest that the trigger of palatalization is not the initial vowel of these suffixes, but different sets of floating melodic features. We provide a formal analysis of the palatalization patterns, as well as of the internal structure of the Czech phoneme inventory, in terms of Element Theory and Strict CV. This allows us to straighforwardly model their lateral (leftward) effect, as well as some peculiar behavior in the context of labials and the lateral /l/. Besides Element Theory and Strict CV, we argue that this analysis provides further support to a substance-free view of phonology, in which phonological representations do not necessarily have a universal, fixed phonetic implementation.

Contents

1 Introduction

Many Czech suffixes trigger palatalization of elements to their left. Some minimal pairs of suffixes involve the same vowel, but differ in that one member of the pair does not, while the other does, trigger palatalization. A case in point are the so-called hard and soft agreement markers, orthographically *-ý* and *-í*, respectively, but both corresponding to the same sound /iː/. While *-ý* does not trigger palatalization of the consonant to its left, *-í* does, as the following pair of adjectives shows.

Table 1: Hard and soft agreement markers

Orthography	Pronunciation	Meaning
mladý	[ˈmladiː]	'young.M.SG'
mladí	[ˈmlaɟiː]	'young.M.ANIM.PL'

The contrast is systematic: *-ý* is never preceded by [ɟ], and *-í* is never preceded by [d], and this complementary distribution extends to the other sounds that can precede these inflection markers (see Sect. 4 below for more discussion of this point). The existence of such contrasts shows that what triggers palatalization is a property of the suffix, and that two suffixes that sound alike can have different palatalization-triggering properties. At the same time, it is the element to the left of the suffix that shows the visible effect of palatalization. This type of pattern is not restricted to Czech, but is characteristic for the Slavic family as a whole.

A further noteworthy fact is that palatalization is not a uniform process. Different palatalizing suffixes trigger different types of palatalization, even though they may begin with the same vowel. The palatalizing suffix *-ě* /ɛ/, for example, which marks the dative singular of feminine nouns, changes /k/ into /ʦ/, while the suffix *-ěn* /ɛn/ of the passive past participle changes /k/ into /ʧ/ (examples provided in Table 2 below). This state of affairs represents another instance of what we claimed in the previous paragraph, namely that two similar sounding objects—e.g. the /ɛ/ of the F.DAT marker and that of the PASS.PTCP—can display different phonological behaviors, as they trigger different types of palatalization.

Our objectives in this paper are twofold. The first is an empirical one. We start off by discussing the two different types of palatalization distinguished in Scheer (2001). We then proceed to show that there are additional patterns on top of the ones discussed by Scheer, such as the ones triggered by the comparative suffix *-ěj*, and the causative suffix *-i*. Our second aim is theoretical. There has been quite a bit of discussion in the literature on the topic of palatalization in Slavic (see Halle 2005, Rubach 2011, Krämer & Urek 2016, and references cited there), which has shaped the debate over phonology and its representations. We will develop an analysis which adopts two main ingredients. The first concerns the particular decomposition of sounds into their component parts, such as it has been proposed in the framework of Element Theory (Backley 2011). The second ingredient we shall need is the strict CV framework familiar from the work of Lowenstamm (1996), Scheer (2004).

2 The data

2.1 Small and big palatalization

As a starting point of our discussion of palatalization in Czech, we describe the data reviewed in Scheer (2001). Scheer distinguishes two types of palatalization, which he labels, following traditional descriptions, 'small' palatalization (also called second palatalization) and 'big' palatalization (or first palatalization). Table 2 lists the two types of palatalization for all the consonants that are liable to being affected by it. In the orthography, the diacritic on the *ě* (widely known under its Czech name *háček* 'little hook') marks the fact that it is a palatalizer.[1] When *ě* is attached to the base, the *háček* sometimes appears on the preceding consonant (as in *že, če, ře*), sometimes on the vowel (in *tě, ně, pě, bě*), and sometimes not at all, as with the labials and /l/, to mark the absence of palatalization.

The first column lists the base sound. The base sounds are ordered from top to bottom according to their place of articulation (velar, coronal, labial). To the right of the first column we find the sounds that result from small palatalization, followed by an example. Small palatalization is indicated by light shading. The rightmost columns show the effect of big palatalization (dark shading). In some cases, there is no difference between small and big palatalization (e.g. /x/, /n/, and /r/ become /ʃ/, /ɲ/, and /r̝/, respectively, in both palatalization patterns). With /s/ and /z/ (and possibly also with /ts/), small palatalization has no visible effect, which is indicated by the absence of any shading. An even more extreme scenario is represented by /l/, which never palatalizes.

The labials present an interesting pattern in a number of respects. Small palatalization does not result in a change in the articulation of the base sound, but

[1] This is different from cases in which the háček occurs on consonants (like š, č, ž, etc.). In such cases, the háček indicates that the consonant itself is palatal.

Table 2: Small and big palatalization in Czech (Scheer 2001)

	small	LOC/DAT.F.SG *-ě*	big	PASS.PTCP *-ěn*
/k/	/ts/	louka–louce 'meadow'	/tʃ/	zatknout–zatčen 'arrested'
/g/	/z/	liga-lize 'league'	/ʒ/	(Olga-Olžin 'of Olga')
/x/	/ʃ/	střecha–střeše 'roof'	/ʃ/	nadchnout–nadšen 'excited'
/ɦ/	/z/	kniha–knize 'book'	/ʒ/	táhnout–tažen 'pulled'
/t/	/c/	máta–mátě 'mint'	/ts/	nutit–nucen 'forced'
/d/	/ɟ/	vláda–vládě 'government'	/z/	uklidit–uklizen 'cleaned up'
/ts/	/ts/	(Bystrica - Bystrice 'Bystrica')	/tʃ/	péci–pečen 'baked'
/s/	/s/	mísa–míse 'bowl'	/ʃ/	hlásit–hlášen 'reported'
/z/	/z/	koza–koze 'goat'	/ʒ/	ohrozit–ohrožen 'endangered'
/n/	/ɲ/	vina–vině 'guilt'	/ɲ/	ranit–raněn 'wounded'
/r/	/r̝/	sestra–sestře 'sister'	/r̝/	(Věra–Věřin 'of Vera')
/l/	/l/	víla-víle 'fairy'	/l/	zvolit–zvolen 'elected'
/p/	/pj/	stoupa–stoup[j]ě 'rise'	/p/	zatopit–zatopen 'flooded'
/b/	/bj/	ryba–ryb[j]ě 'fish'	/b/	rozzlobit–rozzloben 'angered'
/f/	/fj/	karafa–karaf[j]ě 'decanter'	/f/	trefit–trefen 'hit'
/v/	/vj/	hlava–hlav[j]ě 'head'	/v/	obarvit–obarven 'dyed'
/m/	/mɲ/	zima–zim[ɲ]ě 'winter'	/m/	zlomit–zlomen 'broken'

rather in the insertion of a front vocoid between the labial and the suffix. In the case of /m/, this extra segment takes the form of /ɲ/ (Short 1993: 459). Rather counterintuitively, the big palatalization has no effect in the labials, as indicated by the absence of shading. We come to the peculiar behaviour of the labials at the end of this section.

A brief comment is in order on the the underlying representation of the sound affected by palatalization. In the case of small palatalization, the examples show both the nominative and the dative singular forms. The idea that the nominative represents the base sound of the final consonant of the base is well-supported, in that it appears when the base is followed by the non-palatalising suffix *-a*. In the case of big palatalization, however, the nature of the base sound is not always so obvious, because the bases are mostly followed by the theme vowel *-i*, which is itself a (small) palatalizer. For example, in the case of the participle *nuc-en* /nutsɛn/ 'forced', the base is represented by the infinitive *nut-it* /nucit/ 'to force', where the final segment of the base is a palatal /c/. The assumption that the base has an underlying /t/ is supported by the fact that no *i*-verb has a root ending in /t/. This means that, in the case of the suffix *-ěn*, the base sound is often not directly observable.

A further relevant data point concerning the suffix *-ěn* is that its palatalization pattern is not always consistently the same. In Table 2 it is listed as a big palatalizer, but it is possible to identify verbs where *-ěn* shows the pattern of small palatalization. Combined with what we said at the end of the previous paragraph about the infinitive, this leads to a situation where the infinitive and the participle are indistinguishable, both displaying the pattern of small palatalization, e.g. *poctít* /pɔtsciːt/ ∼ *poctěn* /pɔtscɛn/ 'to favour'∼'favoured' and *radit* /raɟit/ ∼ *raděn* /raɟɛn/ 'to advise'∼'advised'. Furthermore, verbs with theme vowel *e* and a root ending in /ts/ have a passive participle with /ts/ (e.g. *kácet*∼*kácen* 'to cut down'∼'cut down'), which would also be consistent with small palatalization of underlying /ts/. Finally, some (athematic) verbs even do not palatalize at all when followed by the passive participle suffix, like *mést*∼*meten* 'to sweep'∼'swept' and *krást*∼*kraden* 'to steal'∼'stolen'. Especially the latter class suggests that, in the case of the suffix *-en*, the vowel and the palatalizer may lead an independent life, as the vowel occurs without the palatalizer. While these facts complicate the empirical picture, we feel that it is safe to set them aside for now, since they do not call into question the existence of the two different types of palatalization that we discuss here. We hope to come back to the more complex patterns discussed in this paragraph in future work.

A final noteworthy fact is the scarcity of data for two sounds, /ɡ/ and /ts/. The first of these appears to be a phoneme that is restricted to loan words (e.g. /liɡa/∼/lize/ 'league') and proper names (e.g. /ɔlɡa/∼/ɔlʒin/), and is otherwise so infrequent in Czech that it cannot be reliably attributed to any of the patterns we discuss. A similar data scarcity problem exists for /ts/. For small palatalization, Scheer lists the Slovak place name *Bystrica*, but otherwise nouns with roots ending in /ts/ appear to be absent from Czech in this particular declension (although a reviewer points out that there are such nouns in other declensions, like *noc* 'night' and *tác* 'plate'). The verbal pair *péci* /pɛːtsi/ ∼ *pečen* /pɛtʃɛn/ 'to bake' may well involve a palatalized /k/, as this sound appears in the past participle *pekl* 'baked' and the noun *pekárna* 'bakery'. This would in effect mean that the apparent /ts/∼/ts/∼/tʃ/ pattern is in effect a /k/∼/ts/∼/tʃ/ pattern. This latter possibility is in fact entirely consistent with the analysis we shall develop. Since data that show the patterns of palatalization for both /ɡ/ and /ts/ are too scarce to base any reliable conclusions on, we shall henceforth ignore both these sounds (although [ts] will make an appearance in the analysis as the result of big palatalization applied to /t/).

2.2 Additional patterns

We now turn to two additional patterns, one involving the comparative suffix *-ějš*, and the other shown by the causative suffix *-i*. Following Caha et al. (2019), we take the comparative suffix to consist of two morphemes *-ěj* and *-š*. We henceforth refer to the suffix as *-ěj*, since this is all we shall need to study the additional palatalization patterns. Table 3 lists the relevant patterns. The shading in this table (and the ones to follow) is the same as in Table 2.[2] We notice that, in terms of the distinction between small and big palatalization that we discussed earlier, comparative *-ěj* shows a mixed pattern: in the velars, it triggers big palatalization, but in the coronals and the labials, we see the pattern typical of small palatalization.

Table 3: Comparative *-ěj*

	base		small		big	
/k/	divoký	'wild'	/ts/		/tʃ/	divočejší
/x/	plachý	'timid'	/ʃ/		/ʃ/	plašejší
/ɦ/	strohý	'austere'	/z/		/ʒ/	strožejší
/t/	kulatý	'round'	/c/	kulatější	/ts/	
/d/	hrdý	'proud'	/ɟ/	hrdější	/z/	
/s/	lysý	'bald'	/s/	lysejší	/ʃ/	
/z/	drzý	'cheeky'	/z/	drzejší	/ʒ/	
/n/	levný	'cheap'	/ɲ/	levnější	/ɲ/	
/r/	bujarý	'merry'	/r̝/	bujařejší	/r̝/	
/l/	rychlý	'fast'	/l/	rychlejší	/l/	
/p/	hloupý	'stupid'	/pj/	hloup[j]ější	/p/	
/b/	blbý	'stupid'	/bj/	blb[j]ější	/b/	
/f/			/fj/	—	/f/	
/v/	hravý	'playful'	/vj/	hrav[j]ější	/v/	
/m/	lakomý	'stingy'	/mɲ/	lakom[ɲ]ější	/m/	

A similar pattern, but with a twist, is found with the causative suffix *-i*. This is shown in Table 4. Since we know that the hard adjectival agreement marker

[2] We put the form *domácí* between brackets since it features the soft declension marker *-í*, and so we cannot be certain that in the case of base-final /ts/ and /z/ we are looking at unpalatalised consonants. The lack of adjectives with the hard *-ý* ending appears to be a gap in the data.

-ý does not trigger palatalization, we use adjectives to represent the base, and restrict ourselves to causative verbs derived from adjectives. Where this was not possible (as with *s* and *z*), we took recourse to nouns. The verbs frequently are accompanied by a prefix, like *z-* or *o-*, and the causative *-i* is followed by the past participle marker *-l*. Again, we see mixed pattern: big palatalization with the velars, and small palatalization with the coronals. The labials now fail to palatalize, which is typical of the pattern of big palatalization.

Table 4: Causative *-i*

	base		small		big	
/k/	trpký	'bitter'	/ts/		/tʃ/	ztrpčil
/x/	tichý	'silent'	/ʃ/		/ʃ/	tišil
/ɦ/	drahý	'expensive'	/z/		/ʒ/	zdražil
/t/	nejistý	'unsure'	/c/	znejistil	/ts/	
/d/	hnědý	'brown'	/ɟ/	zahnědil	/z/	
/s/	kosý	'slanting'	/s/	zkosil	/ʃ/	
/z/	mráz	'frost'	/z/	zmrazil	/ʒ/	
/n/	levný	'cheap'	/ɲ/	zlevnil	/ɲ/	
/r/	moudrý	'wise'	/r̝/	zmoudřil	/r̝/	
/l/	rychlý	'fast'	/l/	zrychlil	/l/	
/p/	tupý	'blunt'	/pj/		/p/	tupil
/b/	chabý	'weak'	/bj/		/b/	ochabil
/f/			/fj/		/f/	
/v/	ošklivý	'ugly'	/vj/		/v/	zošklivil
/m/	chromý	'lame'	/mɲ/		/m/	ochromil

Putting these facts together, it would appear that we end up with four different patterns. These are schematically shown in Table 5. However, we shall argue that the state of affairs depicted in Table 5 results from the interplay of two different factors. The first is the nature of the palatalizer, and the second is the question whether a palatalizer that is underlyingly present surfaces in the output, or remains unassociated. The failure of palatalizers to surface can be observed systematically in labials, in particular with the passive participle suffix *-ěn* and the causative *-i*, where big palatalization has no visible effect. The velars and the coronals clearly show that a suffix-associated palatalizer must be present in the big palatalization pattern, and so the absence of palatalization with the labials has to be due to the fact that something in the UR does not show in the output. We shall call this property Output Invisibility. In Sect. 4.3.3 we shall develop an analysis for Output Invisibility with the labials that relies on the association (or rather, the lack thereof) of elements on the melodic tier (like palatalizers) with CV slots. This allows us to remove the labials from Table 5, and identify the three different palatalization patterns shown in Table 6.

Table 5: Four different patterns

	small	CMPR *-ěj*	CAUS *-i*	big
/k/	/ts/	/tʃ/	/tʃ/	/tʃ/
/x/	/ʃ/	/ʃ/	/ʃ/	/ʃ/
/ɦ/	/z/	/ʒ/	/ʒ/	/ʒ/
/t/	/c/	/c/	/c/	/ts/
/d/	/ɟ/	/ɟ/	/ɟ/	/z/
/s/	/s/	/s/	/s/	/ʃ/
/z/	/z/	/z/	/z/	/ʒ/
/n/	/ɲ/	/ɲ/	/ɲ/	/ɲ/
/r/	/r̝/	/r̝/	/r̝/	/r̝/
/l/	/l/	/l/	/l/	/l/
/p/	/pj/	/pj/	/p/	/p/
/b/	/bj/	/bj/	/b/	/b/
/f/	/fj/	/fj/	/f/	/f/
/v/	/vj/	/vj/	/v/	/v/
/m/	/mɲ/	/mɲ/	/m/	/m/

The three remaining patterns now show the unadulterated effect of the first factor involved, which is the nature of the palatalizer. We accordingly call them small, medium, and big palatalization, and we take them to be triggered by three different types of palatalizers. For now, we simply refer to these palatalizers as PAL_1, PAL_2, PAL_3. Sect. 4 will be devoted to developing an analysis of their internal structure. It is these palatalizers that are associated with certain suffixes, and which trigger the effects on elements to their left.

In the remainder of the paper, we shall make one further simplification of the data, which involves pairs of phonemes that differ with respect to voice, like /x-ɦ/, /t-d/, /s-z/, /p-b/, and /f-v/. This is because, once we have the analysis of the voiceless members of these oppositions in place, adding the voiced member

Table 6: Three different patterns (excluding the labials)

	small PAL$_1$	medium PAL$_2$	big PAL$_3$
/k/	/ʦ/	/ʧ/	/ʧ/
/x/	/ʃ/	/ʃ/	/ʃ/
/ɦ/	/z/	/ʒ/	/ʒ/
/t/	/c/	/c/	/ʦ/
/d/	/ɟ/	/ɟ/	/z/
/s/	/s/	/s/	/ʃ/
/z/	/z/	/z/	/ʒ/
/n/	/ɲ/	/ɲ/	/ɲ/
/r/	/r̝/	/r̝/	/r̝/
/l/	/l/	/l/	/l/

is trivial. That removes the voiced sounds from the table, to wit /ɦ, d, z, b, v/. This will be our starting point for the analysis in the following section.

3 Theoretical prerequisites

In this section, we sketch the theoretical background that we shall be assuming in our analysis, to be developed in the next section. This background involves Element Theory, and strict CV. Before presenting some of the basics of these theories, we devote some general remarks to palatalization, and its previous treatment in the literature.

3.1 Previous approaches to palatalization

Palatalization is "probably the most common phonological assimilatory process, maybe only second to nasal place assimilation" and varies "immensely in its phonological and morphological conditions cross-linguistically". Also from a phonetic perspective, "this phenomenon [...] shows a wide range of variation and it is not quite clear which patterns and processes should be subsumed under palatalization" (Kochetov 2011). Because of this, the formalization of palatalization has triggered a lively theoretical debate, which centered mostly on the properties of the representational component of phonology. Indeed, whereas most researchers agree in analyzing palatalization as feature spreading/sharing, different proposals have been put forward concerning the set of rules and/or constraints involved in the palatalization operation, and the typology of phonological features and their geometric organization (Chomsky & Halle 1968, Clements 1985, Hume 1992, Clements & Hume 1995, Halle et al. 2000, Calabrese 2005, Halle 2005, Morén 2006, Bateman 2007, Kochetov 2011, Krämer & Urek 2016). This disagreement has some relatively far-reaching consequences, as despite the amount of work targeting palatalization, "some of the problems with the formal modeling of palatalization have not been resolved. In part, these difficulties appeared to stem from a more fundamental problem –– the persistent use of traditional [feature-geometric] representations [,] which were assumed to be inviolable, universal, and innate" (Kochetov 2011).

Another representational assumption shared by traditional models is the idea that features are phonetically-grounded. Thus, the objects involved in palatalization (i.e. the trigger and the target) are represented in an extremely phonetically faithful way, there being no slack between the set of phonological features they consist of, and their phonetic implementation. As noted by Kochetov (2011), whereas this has "contributed to [an] empirically adequate account of cross-linguistic patterns of palatalization, it became clear that the same representations have often stood in the way of further empirical coverage of the phenomenon [.] This particularly applies to cases of palatalization that can be considered less phonetically natural, such as place-changing processes resulting in anterior coronals or involving labial consonants." These processes are understood as the spreading of some phonological features from the trigger that add a secondary palatal articulation to the target, followed by a process that changes the place (and the manner) of articulation of the target (and, in approaches based on Chomsky & Halle 1968, by the application of a set of marking conventions). For instance, Chomsky & Halle (1968) formalize the so-called Slavic "first velar palatalization" (/k g x/ → [ʧ ʤ ʃ]) as a clear case of assimilation, where the [−back] feature of the front vowels triggering the palatalization affects preceding velar ([−anterior]) targets, as formalized in (1).

(1) [−ant] → [−back, +cor, +del rel, +strid] / ___ [−cons, −back]

However, the change from stops to coronal affricates and fricatives can be hardly understood as a case of assimilation alone, for front vowels lack the features acquired by the target, namely [+coronal, +delayed release, +strident]. To reach the desired result, the assimilation rule needs to be supplemented with the set of marking conventions in (2), which fills in the unmarked values of underspecified (*u*) features (Kochetov 2011).

(2) a. [*u*cor] → [+cor] / [___, −back, −ant]
b. [*u*del rel] → [+del rel] / [___, −ant, +cor]
c. [*u*strident] → [+strident] / [___, +del rel, +cor]

These conventions imply that the postalveolar affricates and fricatives /ʧ ʤ ʃ/ are less marked than the palatal or palato-alveolar stops /c t̠ʲ/, since the latter lack stridency and delayed release. These marking conventions formally encode the typological observation that "when velar obstruents are fronted, it is simpler for them also to become strident palato-alveolars with delayed release" (Chomsky & Halle 1968: 423). Despite the fact that combining rules and marking conventions allowed a satisfactory level of descriptive adequacy, nothing determined when such marking convention should be applied, nor whether these conventions are appropriate for specific cases of palatalization. For instance, whereas postalveolar affricates are less marked than palatal or postalveolar stops, even less marked are alveolar or labial stops, which are never produced by palatalization. Furthermore, as already mentioned, place-changing palatalization processes would consist of two different processes, namely fronting/secondary palatalization (e.g. /k t/ → /c tʲ/) plus simplification (e.g. /c tʲ/ → /ʧ ʧ/) (see Kochetov 2011: 1677 for further details). The fact that what intuitively looks like a unitary process ("an obvious case of regressive assimilation", Chomsky & Halle 1968: 308) is formalized in terms of two unrelated processes blurs the naturalness the proponents of approaches adopting traditional, phonetically-grounded features were after, and adds to the complexity and the abstractness of the proposed analyses.

Even in approaches that dispense with marking conventions and try to derive the naturalness and (im)possibility of specific process from the geometric properties of featural representations (Clements 1985, Sagey 1988, Hume 1992, Halle 1995, Clements & Hume 1995, Halle et al. 2000, Calabrese 2005, Halle 2005, Morén 2006, among others), similar problems remain. As argued by Kochetov (2011: 1684), even if such approaches further improved on descriptive adequacy of cross-linguistic patterns of palatalization, "some of the processes that could be easily stated in the SPE-style approach [...] could no longer be stated in the feature geometry approach *without making some ad hoc stipulations* (emphasis ours–Author(s))".

Similar problems are inherited by OT-based accounts that more or less explicitly adopt feature geometric representations, the main difference being that in OT-systems the observed variation is explained in terms of constraint ranking. For instance, the relative resistance to palatalization of labials can be formalized as a universally fixed hierarchy of constraints penalizing palatalized labials, dorsals, and coronals (3a) (Chen 1996), and the height asymmetry of triggers as a fixed hierarchy of Palatalize constraints indexed for vowel height (Rubach 2003) (3b).

(3) a. *[lab]/VPl[cor] » *[dors]/VPl[cor], *[cor]/VPl[cor]
    b. Pal/j, Pal/i » Pal/e » Pal/æ

A possible solution to the problems inherited from feature-geometric approaches is using more detailed, phonetically realistic representations. For instance, Chen (1996) combines feature-geometrical representation with articulatory gestures, and analyzes Japanese, Polish, and Swati palatalization patters as all involving the spreading of V-Place[coronal] from front vocoids, which produces abstract complex segments featuring both a secondary place and the original primary place (e.g. [Dorsal]/V-Place[Coronal]). In such system, the observed cross-linguistic variation would derive from a difference in the way these abstract complex segments are phonetically implemented via articulatory gestures. For instance, depending on the language, [Dorsal]/V-Place[Coronal] can be phonetically interpreted as [kʲ], [c], [tʃ], or [tɕ]. This approach, though, is insufficient in accounting for cases where, as in several Slavic languages, one and the same segment (e.g. /k/) undergoes different types of palatalization.

As we show below, our proposal provides a different solution to the aforementioned problems of feature geometric approaches, namely those related to the need to postulate more than one (assimilatory) process, and to cross- and intra-linguistic variation. A crucial component of our solution involves desisting from the attempt of defining a universal featural characterization of segments, which has been proved unsuccessful by a couple of decades of feature-geometrical research (Uffmann 2011). We adopt instead an approach where the internal representation of segments emerges as a function of the segments' phonological behavior, rather than of their phonetic correlates (Mielke 2008, Dresher 2009). We thus abandon phonetically-grounded features and resort instead to a substance-free version of Element Theory (Kaye et al. 1985, Harris 1994, Harris & Lindsey 1995, Backley 2011, Scheer & Kula 2017), which is grounded in a strict approach to modularity (Fodor 1983, Scheer 2012). In such an approach, where phonology and phonetics resort to different lexica, i.e. to different representational primitives, the observed variation can be built into the phonology-phonetics interface: different languages will phonetically interpret one and the same set of elements differently because its phonology-phonetics interface maps the relevant, abstract phonological representation onto the corresponding acoustic form in an arbitrary way (Scheer 2012). Hence, substance-freeness and strict modularity give us a way out from the need of postulating more than one process, as it does not require us to distinguish between a process that spreads a palatalizing feature/element and one that changes the manner specification of the palatalized segment. For instance, this system allows us to model the palatalization of /k/ into [ts] as being due to the spreading of a single feature/element I (see (18)), followed by a lexical-insertion type of phonology-phonetics mapping that arbitrary assigns the abstract /k + I/ representation the acoustic properties of [ts]. Our systems thus makes use of Chen's (1996) hypothesis that the same phonological representation can be phonetically interpreted differently on a language-specific basis, but it also allows for formally modeling the fact that one and the same segment can display different types of palatalization within the same language. As we show below, this follows from the hypothesis that the factor responsible for palatalization is not to be found in the featural make-up of the vowel that follows the palatalized consonant on the surface, but on the presence of different sets of floating elements that associate to the palatalized consonant. By teasing apart audible front vowels and palatalizers we can also provide an account for why one and the same segment undergoes different kinds of palatalization even if, on the surface, it is followed by the same vowel (see e.g. Table 11). In what follows, we introduce and describe in detail the theoretical toolkit we need to develop such system.

3.2 Element Theory

Differently from previous approaches, we dispense with traditional, articulatory-based featural representations and adopt instead a substance-free take on phonology (Reiss 2017, Reiss & Volenec 2022, Scheer 2019, Chabot 2022, 2023), where the internal structure of segments is determined by a segment's language-specific phonological behavior, rather than by its phonetic properties (despite there being a tendency for a correlation between the two, see e.g. Scheer 2015). As we will show below, this allows us to avoid having to decompose place-changing palatalization processes into two formally unrelated processes.

A model that incorporates these properties is Element Theory (Kaye et al. 1985, Harris 1994, Harris & Lindsey 1995, Backley 2011, Scheer & Kula 2017).

In its standard version (Backley 2011), this model assumes the six unary representational primitives in Table 7, where the elements are presented with their acoustic correlate, and the phonological categories they instantiate. Note that the 'Acoustic correlate' and 'Phonological categories' columns of Table 7 list the phonetic and phonological properties that commonly correlate with the relevant elements. This does not mean that these correlations hold universally: given the assumption that the elemental make-up of segments primarily derive from their phonological behavior, and that the phonological behavior of segments is language-specific, there is no reason to expect that segments that belong to a category in the rightmost column of Table 7 will necessarily feature the corresponding element. For instance, as we shall see below (Table 11), we do not assume that all voiceless obstruents contain H, that nasals contain ʔ, or velar consonants contain U.

Table 7: The elements (Backley 2017)

Elements	Acoustic correlate	Phonological categories
A	high F1 (F1–F2 converge)	pharyngeals, coronals, laterals, non-high vowels
I	high F2 (F2–F3 converge)	palatals, coronals, front vowels
U	lowering of all formants	labials, velars, uvulars, rounded vowels
ʔ	abrupt and sustained drop in amplitude	oral/nasal/glottal stops, laryngealised vowels
H	high-frequency energy	fricatives, voiceless obstruents, high tone vowels
L	low-frequency energy	nasals, fully voiced obstruents, low tone vowels

Elements are unary primes that can be used for the representation of both vowels and consonants, and come in the guise of either a head or an non-head, where headedness correlates with greater acoustic and phonological prominence (Backley 2017). Crucially, despite having an acoustic signature, elements are mainly interpreted as cognitive primes reflecting the phonological knowledge of a language user. This means that, during acquisition, the language learner infers the underlying representation of segments from the way they behave phonologically, rather than how they sound. If two segments sound the same, but behave differently, then at a deeper level these two segments are different, i.e. they have two different underlying representations. Thus, we shall expect cases of similar sounding segments that have different underlying representations, and this is exactly what we find in Czech (see below; see also Compton & Dresher 2011 for an example from Inuit).

Following Harris (1994), Botma (2004), we assume that, within a segment, elements are geometrically arranged in a structure featuring a root node (•), which dominates a manner, a place, and a laryngeal node, as depicted in (4). Furthermore, following Kaye et al. (1985) and subsequent work, we assume that each element — e.g. A, I, and U in the place node — sits on its own tier, which, in an approach that further assumes a geometrical arrangement of elements, means that elements are arranged on different tiers in each node too.

(4) Subsegmental structure

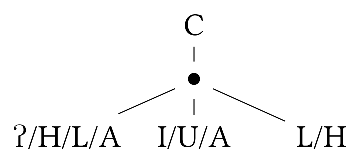
•

[manner] [place] [laryngeal]

In assuming the presence of this root note as a constituent that contains all the features of a segment (Clements 1985), we deviate somewhat from mainstream strict CV. There are several reasons for doing this. One is that we need to be able to refer to sets of elements as units targeted by the phonological computation and by the phonetic interpretation. For instance, the very concept of floating segment, i.e. of a set of elements that is not associated to any C or V slot, would be meaningless if there was no node collecting all the elements it consists of prior to its association with a C/V slot. The need for root nodes in strict CV has recently also been argued for by Ziková (2018), Faust (to appear), and Scheer (2022), who explicitly points out that “timing units (x-slots, Cs and Vs [...], moras etc.) are not the same thing as root nodes [for the] latter have no timing properties but rather define segments” (Scheer 2022: fn. 4).

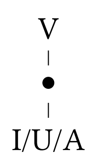
The full set of nodes can be found in consonants, with the manner node hosting ʔ, H, L, or A (or a combination thereof), the place node hosting A, I, or U (or a combination thereof), and the laryngeal node hosting L or H. This is shown in (5), where consonanthood is represented by the root node associating to a C slot. Some elements can be in more than one node. L is used to represent nasals (manner node), as well as voicing and low tones (laryngeal node). Similarly, A is standardly used to represent coronals (place node), as well as laterals (manner node). Given that laterals can have different places of articulation, we believe that they must have both a [manner] and a [place]. Thus, we conclude that A can sit in the manner node. This is in accordance with Backley's proposal given in Table 7 (see Kula 2002 for a proposal featuring the same elements occurring in several nodes).

(5) Consonant structure



Vowels, on the other hand, are traditionally represented with just the place and the root node, the latter associated to a V slot, as in (6). This does not mean that vowels necessarily lack the manner and laryngeal node, and we can indeed find languages that allow for vowels containing the element ʔ (representing laryngealised or creaky quality), or the element L (in nasalised vowels), or, possibly, the element H (representing spread glottis) for devoiced or breathy voiced vowels. As Czech vowels do not show contrasts suggesting the activity of such nodes, we abstain from representing them (unless there is an independent reason for doing so, as in (29) below).

(6) Vowel structure



For ease of exposition in the representations that follow, we shall verticalise the subsegmental geometry given in (4), with the manner node higher than the place one, as shown in (7). Distinguishing manner and place will become important later, when we distinguish phonemes that share the element A, but one has it as a place element (/s/), and another as a manner element (/r/) (see Sect. 4.3.2 for discussion).

(7) Verticalised manner and place

C
|
•
|
[manner]
|
[place]

3.3 Strict CV

The use of C and V to encode the consonanthood or the vowelhood of a set of elements grounds on strict CV (Lowenstamm 1996, Scheer 2004, Passino 2013, Polgárdi 2014, Faust 2014, 2015, 2017, Fathi 2017, Enguehard 2017, Lampitelli 2017, Lahrouchi 2018, Scheer 2019, Newell 2021). Strict CV is a theory of phonological representations that maintains that all phonological strings are sequences of CV skeletal units. More precisely, the X-slots encoding time units in standard autosegmental approaches (Goldsmith 1976) are replaced by skeletal slots encoding both time units and their 'syllabic' status, which can be either consonantal or vocalic, i.e. C or V. These slots strictly alternate: each C slot is followed by a V slot, and each V slot is followed by a C slot, or nothing. No other syllabic constituent is given primitive status.

Thus, what superficially looks like a geminate (8a), or a consonant cluster (8b), will phonologically consist of a CVCV sequence whose left-hand V has no melodic content. The same goes for long vowels (8c) and hiatuses (8d), which are conceived of a CVCV sequence with an internal empty C.

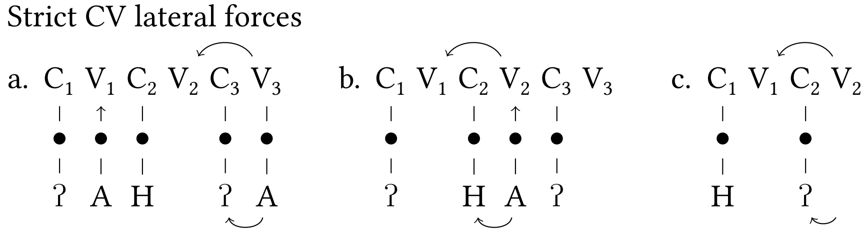
(8) Strict CV structures

a. C V C V	b. C V C V	c. C V C V	d. C V C V
\ /	| |	\ /	| |
•	• •	•	• •
|	| |	|	| |
ʔ	ʔ A	A	I A

As strict CV does not recognize any autonomous/formal status to the syllable (nor to the rhyme), patterns that can be explained with reference to syllabic structure (e.g. open syllable lengthening, closed syllable shortness, extrasyllabicity, coda effects, positional strength, vowel ~ ∅ alternations, etc.), must be given an alternative explanation. Given the flat structure of strict CV representations, this explanation must be found in the lateral relations established between the CV slots. These relations are assumed to be a function of the forces discharged by full or parametrically-licensed (see below) V slots, which always apply leftward and come in two flavors: government and licensing. The former is conceived of as a force that aims at silencing its target. If the target of government is empty, it can stay silent (e.g. no epenthesis), whereas if it contains some melodic structure, such structure is reduced (e.g. lenition). The final V is special in a number of respects. Two separate parameters determine (i) whether it can be empty or not, and, if empty, (ii) whether it can show lateral action, i.e. whether it can govern and license or not. If the first parameter is switched off, the relevant language fills in the final empty V slot via epenthesis, and by virtue of being full, this V slot can govern a preceding empty V slot. If the first parameter is switched on, final empty V are allowed (and words phonetically end in a consonant). In this case, the second parameter determines whether the final empty V slot is a proper governor or not. If it is not a proper governor, we may see vowel-zero alternations at the right edge of the word, as in Czech *loket_* 'elbow.NOM.SG', where the final V slot is empty (indicated by '_') but not a proper governor, so that prefinal V slot cannot be silent, hence the epenthesis of *e*. In contrast, the genitive adds a final vowel *ů*, which governs the prefinal V slot, allowing it to be silent: *lok_tů* 'elbow.GEN.SG'.

Licensing works the other way around, as it supports the pronunciation of the melodic content of its target.[3] These relations are represented in (9), where the leftward arrows on top of the CV tier stand for government, and the ones at the bottom stand for licensing (despite the fact that the bottom arrows link elements on the melody tier, licensing is in fact discharged by V slots on the CV tier).

(9) Strict CV lateral forces



[3] Depending on the specific version of strict CV one adopts, these two basic forces can be upgraded and/or adapted to specific contexts, or reduced to one (see Scheer & Cyran 2018 for an overview).

In a fairly standard version of strict CV (Scheer 2004) the distribution of the two lateral forces depends on three principles: (i) a position cannot at the same time be governed and licensed, (ii) government takes precedence over licensing, and (iii) a full V has to discharge both forces. Government "goes to the preceding empty nucleus, or to the preceding onset if the preceding nucleus is not empty [and] licensing, in turn, affects the position that escaped government" (Scheer & Cyran 2018: 274). Let us see how the system works in (9).

In (9a), the full V_3 scans the string leftward, finds the empty V_2, and discharges its government force on it, while it discharges its licensing force on C_3. Since V_2 is governed by V_3, it can stay silent. By virtue of being governed, it cannot itself discharge any force. Because of this, it cannot govern the empty V_1, which is thus filled in via epenthesis (indicated by the arrow linking the root node of A to V_1). For the same reason, V_2 cannot license C_2, hence the weak status of a position corresponding to traditional codas. In (9b), V_3 is empty. In a language like Czech, where final empty V cannot govern, empty V_2 is ungoverned, and therefore gets filled in via epenthesis (this is the case with the Cz. form *loket* 'elbow' mentioned earlier). Once V_2 gets filled in by epenthesis, it becomes a licit lateral actor, and it can govern V_1, which can remain unpronounced, as well as license C_2. If instead we are dealing with a language where final empty V can be a licit lateral actor (9c), then final V_2 can govern the empty V_1 and license C_2.

The fact that we assume the presence of a root node to which elements of melody are linked implies that we introduce an additional vertical dimension, which generates additional analytical options, which we shall make use of in the analysis that we shall develop below. In the standard case, a C or V slot is associated with a set of melodic elements. This association is mediated by the root node, as shown in (10a): the melodic elements (ʔ or A) are associated with a root node, and the root node is associated with the C or V slot. However, C and V slots can also host no melody, in two different ways, as shown in (10b) and (10c); in (10b), the C or V is associated with a root node but no melody, while in (10c), it is associated with nothing at all. Conversely, (sets of) melodic elements can also lack a C/V host, again in two different ways, as shown in (10d) and (10e). In (10d), the melodic elements are associated to a root node, which itself is floating, i.e. it is not associated with a C or V slot. (10e) represents floating melodic elements, which are neither associated to a root node or a C/V slot.[4]

[4]See Cavirani & van Oostendorp (2017, 2019), Cavirani (2022a) for a further extension of the morpheme template typology, and John (2014), Ziková (2018), Cavirani (2022b), Baturay-Meral & van Oostendorp (2023), Scheer (2022), Faust (to appear) for the need to equip strict CV with the root node.

(10) Strict CV phonological object typology

a.	b.	c.	d.	e.
C/V	C/V	C/V		
•	•		•	
ʔ/A			ʔ/A	ʔ/A

We analyze palatalizers as floating elements, i.e. as cases of (10e). These floating elements are not a primitive of the morphological composition; nor are they part of the base that shows the effects of palatalization. Rather, these floating elements are associated, in the lexical representation, with the suffix triggering the palatalization. Thus, Czech palatalization is a case of featural affixation (Akinlabi 1996, Wolf 2007, Trommer 2012, McPherson 2017). The floating palatalizing elements show visible effects on the final consonant of the base, because, following basic autosegmental assumptions, they strive to associate to the leftmost available slot. The specific details of the mechanism responsible for the association of floating material are rarely spelled out in the literature, an exception being (McCarthy 1981: 382), who maintains that the association of the (floating) elements sitting on the melodic tiers with the skeletal nodes abides by the following association conventions:

(11) i. If there are several unassociated melodic elements and several unassociated melody-bearing elements, the former are associated one-to-one from left to right with the latter.

ii. If all melodic elements are associated and if there are one or more unassociated melody-bearing elements, all of the latter are assigned the melody associated with the melody-bearing element on their immediate left if possible.

As we shall see in Sect. 4.3, these definitions do not exactly cover our case, since the unassociated melodic elements (our floating palatalizers) will need to associate with melody-bearing elements (i.e. the root node of of the base-final C slot) that already contain some element. We believe a minimal reformulation of (11i) will allow this:

i.′ If there are unassociated melodic elements, they are associated with melody-bearing elements one-to-one from left to right.

This formulation will allow that association of floating elements starts at the root node of the base-final C slot, even if that is already filled with melodic material. Association will not be able to reach further leftward into the root because of the ban on crossing association lines. If for some reason association with the final C of the base is impossible (e.g. if it is a labial, see Sect. 4.3.3 for discussion), association will target the root node of the V that follows the base-final C, and so on. If no proper association can be established, elements may remain floating, in which case they are not phonetically realized.

Following John (2014) and Ziková (2018), we further assume that the association of floating melodic elements with C/V slots must always be mediated by a root node. Concretely, a floating element like the one in (10e), with no root node, cannot attach to a C/V slot without a root node as in (10c), as there would be no root node to mediate the association. In order to be able to associate at all, a floating element as in (10e) needs something to its left like (10a) or (10b).

3.4 An alternative scenario

Before proceeding to the actual analysis, there is an alternative scenario that we need to consider (and, as it shall turn out, dismiss). The lateral activity of the palatalizer could also be taken to follow from the fact that all the palatalising suffixes discussed so far involve the front vowel *e* or *i*, both of which contain the element I. This I could be argued to be the factor responsible for palatalization. Under such and a scenario, we would be dealing with a case of (10)a rather than (10)d or (10)e, with I spreading leftward. There are three arguments suggesting that the alternative scenario is incorrect, and that the palatalizer is indeed an additional floating element (or set of elements), as in (10)d or (10)e.

The first argument is that palatalization is not only triggered by morphemes that begin with a front vowel (thus a vowel containing I), but also by morphemes starting with back vowels. Table 8 lists a number of these suffixes, like the suffix *-an*, that derives inhabitants from place names, *-atý*, which derives adjectives, and *-ova*, which is a verbal thematic suffix. A detailed study of suffixes starting with back vowels can be found in Beranová (2009).[5]

A second, more compelling, argument supporting the claim that the palatalizer is independent of the vowel of the suffix is that not all morphemes starting with a front vowel trigger palatalization. For example, in verbs of the *a*-class (which have a theme vowel *-a* preceding the infinitival suffix *-t*), the imperative

[5] Suffixes with back vowels do seem consistently never to trigger glide insertion with labials. We note this fact, but since the particular patterns triggered by back vowels do not concern us here, we put the matter aside.

Table 8: Palatalization triggered by suffixes with back vowels (Beranová 2009)

	base			small		big	
/k/	Africa			/ts/		/ʧ/	Afričan
/x/	ucho	‘ear’		/ʃ/		/ʃ/	ušatý
/ɦ/	Jih			/z/		/ʒ/	Jižan
/t/	Egypt			/c/	Egypťan	/ts/	
/d/	Island			/ɟ/	Islanďan	/z/	
/ts/	Jablonec			/ts/		/ʧ/	Jablonečan
/s/	Persie			/s/		/ʃ/	Peršan
/z/	ohrozit	‘threaten’		/z/		/ʒ/	ohrožovat
/n/	Berlín			/ɲ/	Berlíňan	/ɲ/	
/r/	Katar			/r̝/	Katařan	/r̝/	

is formed by adding the suffix *-ej* to the base. This suffix is minimally different from the comparative *-ěj*, in that it does not trigger palatalization, as is shown in Table 9.

Finally, there is the fact that different morphemes starting with the same front vowel trigger different palatalization patters, as we saw above for the *-ě* suffix of LOC/DAT.F.SG, the CMPR *-ěj* morpheme, and the PASS.PTCP *-ěn* suffix. In each of these three cases, the initial segment *ě* is identical, yet they trigger different palatalization patterns. This fact would seem hard to account for under the simple assumption that the I element of front vowels spreads leftward, triggering palatalization.

The hypothesis that the factor responsible for palatalization is a floating set of elements of features that gets associated with the base-final consonantal segment, rather than the vowel one sees on the surface, is in line with previous accounts of palatalization in Slavic, e.g. Ketner (2005), who proposes an OT account of Czech, Gussmann (1992) and Zdziebko (2015) for Polish, and Bidwell (1962), Iosad & Morén-Duolljá (2010), Padgett (2010) for Russian.

In the next section, we show how the formal tools provided by Element Theory and strict CV allow us to neatly model the Czech palatalization patterns.

Table 9: Imperative *-ej* with *a*-verbs

	INF	IMP	
/k/	čekat	čekej	'wait'
/x/	dýchat	dýchej	'breathe'
/ɦ/	běhat	běhej	'run'
/t/	létat	létej	'fly'
/d/	hledat	hledej	'search'
/ʦ/	kecat	kecej	'piffle'
/s/	klesat	klesej	'decline'
/z/	olízat	olízej	'lick'
/n/	chutnat	chutnej	'taste'
/r/	čmárat	čmárej	'scribble'
/l/	volat	volej	'call'
/p/	čerpat	čerpej	'draw'
/b/	podobat	podobej	'resemble'
/f/	doufat	doufej	'hope'
/v/	dívat	dívej	'watch'
/m/	bloumat	bloumej	'wander'

4 Analysis

4.1 Overview

As we mentioned in the previous section, Czech displays cases of similar sounding segments that have different underlying representations, thereby providing support to a substance-free take on phonology. Czech thus provides support also for the hypothesis that "featural representations [...] possibly reflect local generalizations, specific to certain morphological domains or lexical strata (as, for example, in *cases of multiple palatalization processes targeting the same consonants*). However, these and many other implications for analyses of palatalization have not yet been explored" (Kochetov 2011; emphasis ours–EC & GVW). In this paper, we do explore these implications and provide a formal modeling of the different palatalization processes targeting Czech consonants, showing that the differences in the degree a consonant is palatalized is a function of its underlying representation interacting with that of the exponents of different morphosyntactic objects.

Our proposal for the elemental composition of the three palatalizers is summarized in Table 10, where X is a variable standing for whatever the element composition is of the plain, i.e. unpalatalized, consonant. PAL_1 adds the element I to the plain sound, PAL_2 adds H.I, and PAL_3 adds H.$\underline{\mathrm{I}}$. We use the ∪ symbol to indicate a property of the phonetic interpretation of the combination of the elements of the palatalizers with those of the preceding segments, which is that they compose in the manner of members of a set, which can only contain a single occurrence of the same member, i.e. $\{a, a\} = \{a\}$ (Postma 2019: 11). In the same manner, adding H or I to a node that already contains it, does not add anything. We use the ∪ notation in a purely descriptive manner, without wanting to imply that there is formal operation of set union involved. For our purposes it suffices that we assume that for the process of phonetic interpretation, multiple occurrences in the same C or V slot of the same element below the place, manner, or laryngeal node count as one. The reason for this is that adding a specific element to a node of a segment that already contains that element has no phonological nor phonetic consequence.[6]

Table 10: The elemental composition of the three palatalizers

plain	PAL_1 adds I	PAL_2 adds H.I	PAL_3 adds H.$\underline{\mathrm{I}}$
X	X ∪ I	X ∪ H.I	X ∪ H.$\underline{\mathrm{I}}$

The consonants featuring in the small, medium, and big palatalization patterns (see Table 6 above) have the elements of their plain alternant (represented as X in Table 10), plus those of the relevant palatalizer: I for PAL_1, H.I for PAL_2, and H.$\underline{\mathrm{I}}$ for PAL_3. The element-theoretic representations we shall provide will allow us to develop an analysis where the apparently irregular picture that appears from the phonetics of the palatalized consonants (summarized in Table 6) gets fully regularized.

As can be noted, the palatalizers are in a containment relation. PAL_1 contains only I, which represents frontness; PAL_2 contains I, plus H, the element that represents frication (in the case of fricatives) or audible releasie (in the case

[6]Elements like A, which can be a place or manner element, or L/H, which can be manner or laryngeal, can of course combine in the same segment, provided that they belong to different nodes.

of stops); and PAL_3 contains H, plus the headed version of I, which stands for palatality (Backley 2011).[7] This relation, coupled with the resulting incremental representational complexity, formally translates the degrees or strengths of palatalization we discussed above. As we shall see below, the decomposition of the palatalizers into two distinct atoms H and I also explains the fact that palatalization triggers a change in both the place and the manner of articulation of the targeted consonant (see Ketner 2005 for an account of Czech palatalization in terms of assibilation).

We now proceed to giving overviews of the elemental makeup of individual sets of sounds, starting with the velars and coronals, shown in Table 11. In our tables we adopt the convention that the elements relating to manner are separated by a dot from the elements relating to place.

Table 11: Velar and coronal plain and palatalized consonants

plain		PAL_1 adds I		PAL_2 adds H.I		PAL_3 adds H.$\underline{\text{I}}$	
/k/	ʔ	/ts/	ʔ ∪ I	/tʃ/	ʔ ∪ H.I	/tʃ/	ʔ ∪ H.$\underline{\text{I}}$
/x/	H	/ʃ/	H ∪ I	/ʃ/	H ∪ H.I	/ʃ/	H ∪ H.$\underline{\text{I}}$
/t/	ʔ.A	/c/	ʔ.A ∪ I	/c/	ʔ.A ∪ H.I	/ts/	ʔ.A ∪ H.$\underline{\text{I}}$
/s/	H.AI	/s/	H.AI ∪ I	/s/	H.AI ∪ H.I	/ʃ/	H.AI ∪ H.$\underline{\text{I}}$
/n/	L.A	/ɲ/	L.A ∪ I	/ɲ/	L.A ∪ H.I	/ɲ/	L.A ∪ H.$\underline{\text{I}}$
/r/	A	/r̝/	A ∪ I	/r̝/	A ∪ H.I	/r̝/	A ∪ H.$\underline{\text{I}}$

The two leftmost columns show the IPA symbol and the elemental composition of the plain alternants, while the other columns show, from left to right, the IPA symbol and elemental composition of the small, medium, and big palatalized alternants. The palatalized alternants consistently feature the elements of the corresponding plain alternant, plus those of the relevant palatalizer.

[7] Applying the containment logic to headed $\underline{\text{I}}$ requires that we decompose it into an (unheaded) element I, which it contains, and a factor 'headedness'. One way of conceiving of the headedness factor is that it involves the addition of a second identical element, not in an operation of set union, but one of addition, e.g. $\underline{\text{I}}$ = I + I. We refrain from a more precise formalisation of this idea in the present context, since it the issue is orthogonal to our concerns, and refer the reader to Kula (2002) for a different proposal developing a structural account of headedness, and Backley (2017), Scheer & Kula (2017) for a recent discussion of headedness in Element Theory.

Detailed representations of the Czech morphemes containing the palatalizers will be given in Sect. 4.2. As for the representation of the plain alternants of the base-final consonants (given in the second column of Table 11), we adopt a relatively conservative approach (see e.g. Gussmann 2007, Cyran 2010, Zdziebko 2015, 2022 for comparable element-theory-based representation of other Slavic languages). We take the velar consonants to be underspecified for place of articulation, i.e. to have no place element. This is in line with most previous work on Slavic (Harris & Lindsey 1995, Rice 1996, Gussmann 2007, Cyran 2010, Zdziebko 2015, 2022), and possibly explains why, upon palatalization, velars shift their place of articulation so easily.[8] Coronal consonants contain the place elements A and/or I, which are both traditionally used for coronality (Backley 2011). Coronal stops contain the manner element ʔ, and fricatives H. Nasal and rhotic coronal consonants are represented as L.A and A, respectively.

Voiced obstruents are not shown in Table 11 and subsequent tables. This is because their representation is entirely predictable on the basis of their voiceless counterparts, to which they are identical, except that they add the element $\underline{\text{L}}$ representing voicing in the laryngeal node. For example, /d/ would be represented as ʔ.A.$\underline{\text{L}}$, and its palatalized alternants would be ʔ.AI.$\underline{\text{L}}$, ʔH.AI.$\underline{\text{L}}$, and ʔH.A$\underline{\text{I}}$.$\underline{\text{L}}$, respectively, and similarly for the other obstruents (see Backley 2017 for arguments supporting the need for expressions containing multiple heads). Another sound that is missing from Table 11 is the lateral approximant /l/. It is subject to the exceptionless generalisation that it never undergoes palatalization in Czech. We return to this sound in Sect. 4.3.4 below.

We next present the overview of the labials in Table 12. As discussed above, labial consonants behave differently from other consonants with respect to palatalization. First, the palatalizers never manage to fully palatalize them, i.e. labials never change their place of articulation, nor their manner of articulation. Second, the palatalizers can surface autonomously. As Table 12 shows, PAL_1 and the PAL_2 of CMPR surface after the labial consonant as "a fully developed palatal

[8] An alternative view is put forward by Scheer (2004), who argues that velars contain U, while labial consonants contain an element B (for labiality/roundness). His argument is based on the behaviour of the Czech vocative suffix, which surfaces as [u] after velars, [i] after palatals, and [ɛ] after dentals and labials, where the first allomorph in particular suggests the presence of U in velars. Scheer's proposal is not incompatible with our analysis, provided that (i) we revise the representation of labials and assign them $\underline{\text{U}}$, rather than U, and (ii) assume that $\underline{\text{U}}$ repels the I element of palatalizers, whereas the U element of velars (being weaker than $\underline{\text{U}}$) is 'pushed out' by I. We will not decide here between these alternatives, and continue to use the system of Table 11.

element”, i.e. a front vocoid (Short 1993: 459); we shall argue in 4.3.3 that it is actually a vowel.

Another peculiarity (which seems to be specific to Czech) is that “after bilabial /m/ nasal resonance extends over both segments”, resulting into [mɲ] (Short 1993: 459).[9] This is shown in the PAL$_1$ and CMPR *-ěj* columns of Table 12.

The bisegmental nature of the palatalization is indicated by the n-dash in the representations (e.g. ʔ.U–I). The Output Invisibility of PAL$_2$ of CAUS and PAL$_3$ is indicated by putting the palataliser between brackets (e.g. ʔH.U–(I) in the CAUS *-i* column on the first row). As we argue below, the Output Invisibility of the palatalizer with labials is not related to the strength of the palatalizer, but to the skeletal profile of the suffix, more specifically whether or not the inital vowel of the suffix is itself associated with a V slot. The table thus brings out the fact that, despite there being four columns or patterns of palatalization, there are only three different palatalizers.

Table 12: Labial plain and palatalized consonants

plain		PAL$_1$ adds I		PAL$_2$ adds H.I				PAL$_3$ adds H.I̲	
				CMPR *-ěj*		CAUS *-i*			
/p/	ʔ.U	/pi/	ʔ.U–I	/pi/	ʔH.U–I	/p/	ʔH.U–(I)	/p/	ʔH.U–(I̲)
/f/	H.U	/fi/	H.U–I	/fi/	HH.U–I	/f/	HH.U–(I)	/f/	HH.U–(I̲)
/m/	L.U	/mɲ/	L.U-L.I	/mɲ/	LH.U-I	/m/	LH.U–(I)	/m/	LH.U–(I̲)

The behavior of Czech labial consonants fits with the palatalization typology described by e.g. Bateman (2007), who shows that labials are the consonants that are least affected by palatalization, and when they do, they tend to acquire a secondary place of articulation, rather than changing their manner of articulation (Bateman 2007: 47). As a matter of fact, she observes that labial consonants “never show full palatalization in purely phonological contexts [and] when labials do appear to show full palatalization, this is the result of historical changes”

[9] See Morén (2003) for a comparable phenomenon in Serbian, where all labials show the introduction of some manner feature on an independently realized palatalizer, which surfaces as [ʎ]. The same happens in Russian, where the palatalizer following labial consonants surfaces as [l^{j}] (Halle 1959).

(Bateman 2007: 85). Bateman (2007: 47) explains these facts referring to the articulation: "phonological full palatalization of labials is not expected to occur because of the articulators involved in producing labial consonants on the one hand and the palatalization triggers on the other: the lips and the tongue, respectively. These articulators can move independently of one another, therefore there is no pressure for full palatalization to occur".[10] As we discuss later (Sect. 4.3.3), an autosegmental phonology-based account of the failure of labial consonants to palatalize can be developed that derives such failure from the assumption that, in some languages (e.g. the languages that do not have front rounded vowels), the elements I and U are unable to merge to create a complex place IU, an impossibility standardly attributed to the hypothesis that, in those languages, I and U sit on the same tier (Kaye et al. 1985, Postma 2019). In Czech, the ill-formedness of IU is shown by the fact that Czech does not have front rounded vowels (/y/; Janků 2022), and does not allow for palatalization of labials, suggesting that I and U cannot be combined, which in turn suggests that, in the place node, the tiers of U and I are conflated into one. Because of this, they are in competition, and a Czech well-formed representation can feature only one of them.

In sum, we maintain that an approach that exploits the representational technology of Element Theory and strict CV allows us to provide a neat account of the palatalization patterns of labial consonants too, to derive (i) why labials cannot be fully palatalized, (ii) why the palatalizers can fail to surface, and (iii) when they do surface, the different ways the palatalizers surface, i.e. as a vowel, or as a palatalized nasal.

Before showing the specifics of the analysis of the different palatalization patterns of Czech, and introducing the relevant morphemes and their representation, we expound the logic we adopted for establishing the elemental structure of Czech segments, palatalizers included. As we mentioned above, we assume standard Element Theory, which gives us a set of 6 elements, augmented by the head *vs* non-head distinction. Assuming that this (plus the C/V given by strict CV and the subsegmental geometrical structure) is all language learners have at their disposal, we maintain that language learners start building their phonological representations by mapping the elements onto the acoustic signature the elements encode by default. Crucially, if the phonological behavior gives

[10] This is not to rule out the existence of palatal labials, as in Polish, where Polish, where bases can end in either a plain or a palatalized labial, the difference showing in the fact that they may be followed by the same suffix and be pronounced differently (T. Scheer, p.c.). In such cases, the palatalization is the addition of a palatal offglide [^{j}], and no full palatalization involving a change in the place and/or the manner of the palatalizee.

the learners reasons for doing that, they modify the previously established mappings, and update their phonological representations (Hamann 2009, 2010, Cyran 2012, Scheer 2014, Cavirani 2015).[11] In reconstructing the phonological representations of Czech consonants and of the palatalizers, we followed the same procedure, namely, we started from commonly assumed element-acoustic signature mappings (see Backley 2011 and Table 7 for standard mappings, and Zdziebko 2015, 2022 for previously proposed mappings holding for Slavic languages), and we updated the resulting representations when required by the phonology, i.e., in our case, by palatalization patterns. An important guiding principle we adopted was building the system (i.e. the union of Tables 11 and 12) in such a way that there is no underlying representation that corresponds to several phonetic interpretations. We maintain that this follows if we assume that, at the phonology-phonetics interface, a given phonological object is mapped onto the corresponding abstract acoustic object (akin to Hermann Paul's sound image), in the same way a concept is mapped onto the corresponding phonological form (Boersma 2009, 2011, Hamann 2010, Cavirani 2015, Scheer 2022).[12] For instance, we expect one and the same grammar to map the element A associated to a V node to its

[11]Hamann (2009: 122) maintains that in order to acquire these mappings, language learners pass trough two stages: they identify a few salient acoustic cues (e.g. energy peaks), "keep track of the statistical distribution of [these] cues" and "construct language-specific categories". In the second stage, the lexicon guides the learners to acquire "the labels for the learnt phonetic categories", i.e. the elements that make up phonological representations. Thus, in this stage, the language learners learn which piece of acoustic information is phonologically relevant. Assuming an arbitrary phonetics-phonology interface, we cannot exclude a language where, e.g., A, I, and U are cued, respectively, by [i], [u] and [a] (Scheer 2014). A system such as this, though, would hardly survive inter-generation transmission, as the learner would need other (than acoustic) cues to identify the elements making up a vowel, i.e. to understand "that what they hear is not what they need to store". In case these 'alternative' cues are lacking, the learner would presumably 'store what she hears'.

[12]Following the BiPhon model architecture (Boersma 2011), we assume that the abstract acoustic object a surface phonological representation is mapped onto is in turn mapped onto an articulatory form, i.e. the set of instructions given to the motor system to produce the relevant sound. We argue that the small acoustic differences in the realization of a specific phonological object by one and the same grammar—e.g. the higher probability of English /uː/ to be diphtongized in *move* than in *truth*—arise in this mapping process, and are thus outside of the purview of the phonological grammar. This in no way excludes the possibility that a phonological object can be mapped onto different sound images, but we maintain that this is true only from a typological perspective, namely, different grammars can map representationally identical phonological objects onto different acoustic object, as the mapping is (partially) acquired, thus language-specific. See e.g. Chen (1996), who represents palatalized velars as [Dorsal]/V-Place[Coronal], which, depending on the language, is phonetically interpreted as [kj], [c], [ʧ], or [ʨ].

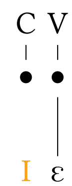
sound image, i.e. [a]. In the same way, in English, the concept CAT is mapped onto /cæt/. What we do not expect, is for one and the same grammar to map the element A̲ associated with a V node to both [a] and [u], in the same way as we do not expect it to map CAT onto both /kæt/ and /dɒg/. The opposite scenario, though, namely one where one and the same acoustic object corresponds to different underlying representations, is not problematic. For instance, in English the string [kæt] can realize both a noun and a verb (e.g. *to cat around*), i.e. can have different morphosyntactic representation. In the same way, [ə] can be the realization of the element A associated with a V node, as well as of the enhanced release of a plosive (Cavirani 2015, Hall 2006, Ridouane & Cooper-Leavitt 2019, Hamann & Miatto 2024), which in some Element Theory-based analysis is represented by the element H (Backley 2011). In such cases, the difference between the two similar-sounding objects would be signaled by a difference in phonological behavior (Compton & Dresher 2011, Hamann 2015), just as the difference between the noun and the verb *cat* is signaled by a difference in morphosyntactic distribution.

4.2 The palatalizers

Among the morphemes triggering small palatalization (PAL_1) we have the suffix expressing the locative and dative of feminine nouns in the singular: *-ě*. Its representation is given in (12) (for ease of exposition, we do not show the elemental composition of /ɛ/ in this and subsequent representations, unless necessary). As discussed above, we argue that the factor responsible for palatalization is not the audible vowel /ɛ/, but a floating place element I that belongs to the same morpheme. We also maintain that the initial C slot of *-ě*.LOC/DAT.F.SG, as well as of other suffixes, is associated to an empty rood node, which will serve as the host for the spreading of the L element of the base-final /m/ (27) and the A element of the base-final /l/ (32).[13]

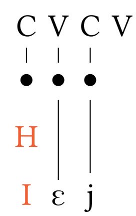
[13]More precisely, it associates to the empty place node under the root node. We ignore this complication for ease of exposition.

(12) *-ě*.LOC/DAT.F.SG - PAL$_1$



As an example of PAL$_2$, in (13) we give the morpheme of the comparative suffix, i.e. *-ěj*. In this case, too, the factor responsible for palatalization is not the audible /ɛ/, but some floating element. Differently from PAL$_1$, we argue that PAL$_2$ contains two such floating elements, i.e. the manner element H and the place element I. Given that we assume strict CV, this morpheme (and the one for PAL$_3$ below) ends with an empty V slot (we remain agnostic on whether this final V slot contains a root node or not).

(13) *-ěj*.CMPR - PAL$_2$

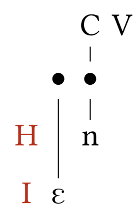


Another PAL$_2$ morpheme we will make use of in the derivations below is the suffix expressing causativity: *-i*. As *-ěj*.CMPR, it consists of the two floating elements H and I. However, differently from *-ěj*.CMPR, these are followed by a floating segment. The latter consists of a root node associated with I, and surfaces as /i/ when associated to a V slot. Its representation is shown in (14). As we shall show below (Sect. 4.3.3), this representation allows us to derive Output Invisibility, namely the failure of the palatalizer to surface independently when following a labial consonant.

(14) *-i*.CAUS

PAL$_3$ is exemplified by the morpheme lexicalising PASS.PTCP *-ěn*, whose representation is shown in (15). PAL$_3$ is a stronger version of PAL$_2$, as its I element is headed, while in PAL$_2$ it is unheaded. Like in the previous morphemes, the palatalizers are floating elements. Like causative *-i*, the palatalizer PASS.PTCP *-ěn* morpheme shows Output Invisibility following labials. We take this to mean that the elements of the audible vowel /ɛ/, i.e. AI, are associated to a floating root node, as was the case with causative *-i* (see (14) above). Details of how this works will be provided below (Sect. 4.3.3).

(15) *-ěn*.PASS.PTCP - PAL$_3$



Note that this morpheme is potentially decomposable into two morphemes, namely H.I plus /ɛ/ on the one hand, and /n/ on the other. An argument supporting this decomposition can be found in the fact that in verbs of the *a*-class, the PASS.PTCP morpheme surfaces as *-an*, whereas in other verb classes, *n* is always preceded by *e*, e.g. *čekat-čekán* 'to wait' vs *sušit-sušen* 'to dry', *sedět-seděn* 'to sit'.[14] We take this to suggest that the *a* theme vowel takes the same position as *ě*, and that the *e* and *i* theme vowels are outcompeted by *ě*. As this has no effect on the palatalization patterns this paper focuses on, we do not explore this issue further.

4.3 Derivations

In this section, we describe in detail the derivation of the palatalization patterns discussed in Section 2. We start from the derivation of palatalized velars (Section 4.3.1), which is followed by the derivation of palatalized coronals (Section 4.3.2) and labials (Section 4.3.3). The derivation of the (failed) palatalization of /l/ is described in a dedicated section at the end (Section 4.3.4).

As already anticipated, three crucial components of our analysis are the hypotheses that (i) the palatalizers are floating elements that look for an available

[14]We thank Ora Matushansky for pointing out this fact, which holds for Russian, but also for other Slavic languages, Czech included.

host, (ii) the appropriate hosts for floating elements are root nodes (, more precisely, the manner and place nodes they contain), and (iii) floating elements look for an available root node left to right (see (11)). In what follows, we show how such hypotheses allows us to model the whole set of palatalization patterns discussed in Section 2.

4.3.1 Velar consonants

We begin our detailed discussion of the derivation by looking at how velar consonants are modified by the three kinds of palatalizers we discussed in the previous section, namely *-ě*.LOC/DAT.F.SG (PAL_1), *-ěj*.CMPR (PAL_2), and *-ěn*.PASS.PTCP. Since *-i*.CAUS (PAL_3) will become relevant only with labial consonants, we will ignore it in this section.

What we need to derive are the palatalized alternants of /k x/, whose representations and IPA transcriptions are given in Table 13 (which in part repeats Table 11 above). For reason of space, we only give the full derivation of the palatalized alternants of the velar stop /k/. The derivation of the velar fricative would be identical to that of /k/, provided that one replaces ʔ with H. Similarly, the derivation of the voiced velar consonants would be identical to that of their voiceless counterparts, provided that one adds the L̲ element representing voicing in the laryngeal node.

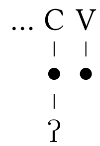
Table 13: Velar plain and palatalized consonants

plain		PAL_1 adds I		PAL_2 adds H.I		PAL_3 adds H.I̲	
/k/	ʔ	/ts/	ʔ ∪ I	/ʧ/	ʔ ∪ H.I	/ʧ/	ʔ ∪ H.I̲
/x/	H	/ʃ/	H ∪ I	/ʃ/	H ∪ H.I	/ʃ/	H ∪ H.I̲

In (16) we give the representation of a base ending with a velar stop. What precedes /k/ is irrelevant, and we represent it by three dots.[15] We assume that silent V slots occurring at the end of a base display a root node with no element attached. In certain cases (to be discussed later on), the floating palatalizer will be shown to associate with this V, and surface as the vocalic segment /i/. In other cases, it will serve as the host for the floating IA and I that we have postulated in the suffixes *-ěn*.PASS.PTCP and *-i*.CAUS, respectively.

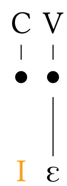
[15] As shown in e.g. Scheer (2001), palatalization can spread backwards in some consonant clusters. As this has no consequence on our reasoning, we do not address such forms.

(16) /k/-final bases



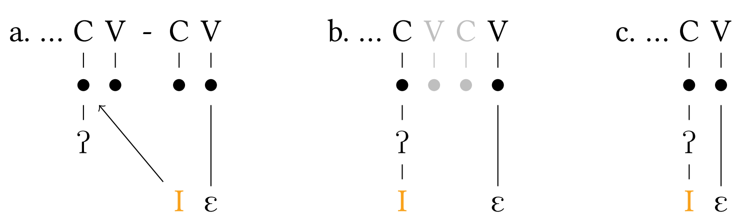
As an example of PAL_1, in (17), we repeat the morpheme expressing locative and dative case in feminine singular nouns, already presented in (12).

(17) -*ě*.LOC/DAT.F.SG - PAL_1



When a base ending in /k/ is concatenated with a PAL_1 morpheme such as -*ě*.LOC/DAT.F.SG, the floating palatalizer tries to associate to the leftmost object it can without crossing association lines. As can be seen in (18), the leftmost available object is the root node containing /k/, i.e. the placeless ʔ. Root nodes to the left of the one containing /k/ are out of sight due to the presence of an association line associating ʔ to its root node. Thus, the palatalizer associates to the latter (18a). This results in a consonantal segment containing ʔ in the manner node, and I in the place node (18b), which is interpreted by the phonetic module as [ts]. As shown in (18b), we assume that a reduction operation applies, which deletes a sequence of empty VC slots (in light gray) (Gussmann & Kaye 1993). This results in the surface representation in (18c).

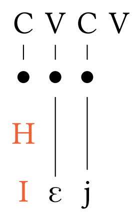
(18) k-ě → [tsɛ]



Let us now see how to derive the second palatalized alternant of /k/, i.e. [ʧ].

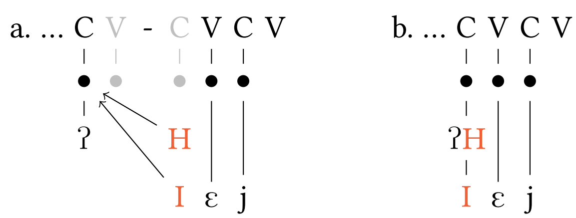
As discussed above, a morpheme containing PAL_2 is *-ěj*.CMPR in (19) (repeated from (13)).

(19) *-ěj*.CMPR - PAL_2



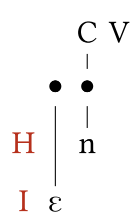
As we saw in the previous case, when concatenated with a base, the floating palatalizer elements contained in *-ěj*.CMPR scan the phonological string from left to right looking for an available root node. In this case as well, the first available root node is the one containing ʔ. The palatalizers H and I associate to it (20a), and VC reduction applies, yielding a root node containing the segment ʔH.I (20b), which is interpreted by the phonetics module as [ʧ].

(20) k-ěj → [ʧɛj]



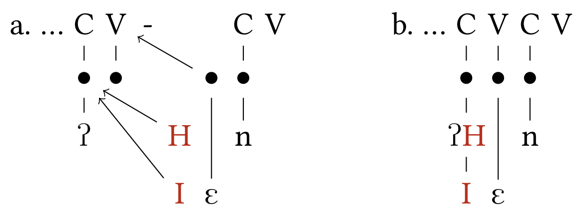
Finally, we look at what happens when a base ending in /k/ concatenates with a morpheme bearing PAL_3, like *-ěn*.PASS.PTCP in (21) (repeated from (15)).

(21) *-ěn*.PASS.PTCP - PAL_3



In this case, we need to accommodate the two floating palatalizer elements—palatalizer H and I̲—and the floating segment /ɛ/. As in the preceding examples, upon concatenation, floating elements look for an available root node scanning the phonological string left to right. The floating H and I̲ find the root node containing ʔ and associate to it. The next floating object, i.e. the floating segment /ɛ/, follows the same procedure: it scans the phonological string left to right looking for an available host to associate to. Differently from floating elements without a root node, floating root nodes look for an available C/V slot. Associating with the C slot containing ʔ would imply crossing the association line linking the base-final V node to its root node. Hence, the floating root node containing /ɛ/ associates to the base-final V slot (22a). This yields (22b), which features ʔH.I̲ in the base-final C slot and /ɛ/ in the base-final V. The phonetic module interprets this as [ʧɛn].

(22) k-ěn → [ʧɛn]



The reader may note that ʔH.I and ʔH.I̲ are interpreted identically—i.e. as [ʧɛn]—by the phonetic module. A similar situation is found with the fricative /ʃ/, which corresponds to four different underlying representations. First, it is the outcome of small, medium and big palatalization of velar /x/. The latter is represented by the sole element H, which turns into [ʃ] when merged with PAL_1 (I), PAL_2 (H.I), and PAL_3 (H.I̲). Furthermore, [ʃ] is also the outcome of big palatalisation applied to /s/, in which guise it corresponds to H.AI̲. Thus, four different underlying representations have the same phonetic interpretation. We show these for convenience in Table 14. The union operation on the relevant elements (applied at the level of phonetic interpretation) will in fact obscure the difference between small and medium palatalisation for /x/ and /s/.[16]

[16] A reviewer asks what would be the UR of /ʃ/ in words like Cz. [ʃ]*akal* 'jackal' and [ʃ]*um* 'noise', where [ʃ] is not the result of a palatalization process. Its UR would minimally have to include H.I, possibly with an added place element A, or headedness for I. It is unclear to us what evidence would allow us to decide between these options. Since this matter is orthogonal to our

Table 14: Four representations for /ʃ/

plain		PAL$_1$		PAL$_2$		PAL$_3$	
/x/	H	/ʃ/	H $\cup$ I	/ʃ/	H $\cup$ H.I	/ʃ/	H $\cup$ H.$\underline{\text{I}}$
/s/	H.AI	/s/	H.AI $\cup$ I	/s/	H.AI $\cup$ H.I	/ʃ/	H.AI $\cup$ H.$\underline{\text{I}}$

As we discussed in Section 4.1, we maintain that this is not a problem if one assumes that the phonology-phonetics interface works via lexical access (Scheer 2014, 2022, Hamann 2010, Boersma 2011): Czech has one palatalized alternant of /x/, i.e. [ʃ], which is generated when merging the 'weakest' palatalizer, i.e. PAL$_1$. Given that there is no other palatalized alternant of /x/ available in the Czech phonetic lexicon, the phonetic module will always map the resulting structure onto [ʃ], even if we merge stronger palatalizers. Returning to the velars, something similar holds for /k/. In this case, however, the Czech phonetic lexicon has two palatalized alternants: [t͡s] (PAL$_1$) and [t͡ʃ] (PAL$_2$ and PAL$_3$).

4.3.2 Coronal consonants

The set of plain and palatalized coronal consonants is given in Table 15 (which partly repeats Table 11 above). As in the case of velar consonants, the three degrees of palatalization consistently show I, H.I, and H.$\underline{\text{I}}$, respectively.[17]

The derivation of coronal consonants is identical to that of the velar consonants: upon concatenation, the floating palatalizers contained in the suffix scan the phonological string left to right, and associate to the leftmost available root node, i.e. the one associated to the base-final C slot, thereby generating the palatalized alternants given in the PAL$_1$, PAL$_2$, and PAL$_3$ columns in Table 15. The latter are then interpreted by the phonetic module in the same fashion we saw above, namely via lexical access. As discussed above, this allows for the presence of similar-sounding phones corresponding to different underlying representations.

Despite the fact that the derivation of the palatalized alternants of the coronal consonants adds nothing to the mechanics that we saw earlier with the velars,

concerns, we shall leave it aside, referring to Kula (2008) for relevant discussion with reference to palatalization in Kinyamwezi.

[17]In principle, Table 15 should feature lateral /l/ too. However, its behavior is atypical for coronals, in that it never shows any palatalization. This property makes it resemble the labials, which may also fail to show palatalization in the presence of palatalizers. Lateral /l/ will receive a special treatment in Sect. 4.3.4 below.

Table 15: Coronal plain and palatalized consonants

plain		PAL$_1$ adds I		PAL$_2$ adds H.I		PAL$_3$ adds H.I̠	
/t/	ʔ.A	/c/	ʔ.A ∪ I	/c/	ʔ.A ∪ H.I	/ʦ/	ʔȦ ∪ H.I̠
/s/	H.AI	/s/	H.AI ∪ I	/s/	H.AI ∪ H.I	/ʃ/	H.AI ∪ H.I̠
/n/	L.A	/ɲ/	L.A ∪ I	/ɲ/	L.A ∪ H.I	/ɲ/	L.A ∪ H.I̠
/r/	A	/r̝/	A ∪ I	/r̝/	A ∪ H.I	/r̝/	A ∪ H.I̠

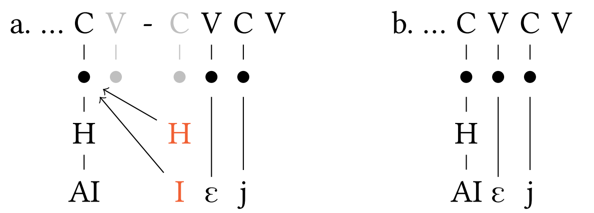
we want to show the derivation of the palatalized alternants of /s/ and /r/, as this gives us a chance to show the role played by the subsegmental geometry we gave in (7), repeated here as (23).

(23) Verticalised manner and place

C
|
•
|
[manner]
|
[place]

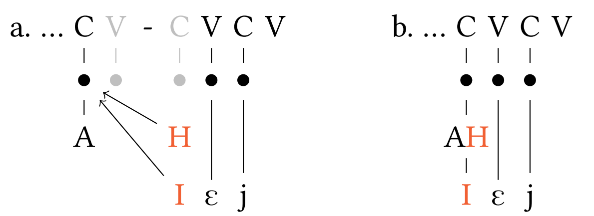
As shown in Table 15, the PAL$_1$ and PAL$_2$ alternants of /s/ and the PAL$_2$ alternant of /r/ display the same set of elements: H, A, and I. The same holds for the PAL$_3$ alternants of /s/ and /r/, which both display H, A, and I̠. Despite the fact that these elements sets look identical, they differ in terms of internal structure. In their plain alternant, /s/ has the element H in the manner node and the elements AI in the place node (represented as H.AI), while /r/ has the A element only, which we assume is sitting in the manner node (in /r/ as well as in /l/; see Sect. 4.3.4). When merged with the palatalizers, the difference between these two structures is maintained. This is shown in the compact notation by the position of the dot: palatalized /s/ has H.AI (or H.AI̠), whereas platalized /r/ has AH.I (or AH.I̠). The representations in (24) and (25) represent this graphically. In (24a) we have a base ending in /s/ that concatenates with *-ěj*.CMPR. As shown, the root node of the base-final C slot of the output form (24b) contains the element H in the manner node and the elements AI in the place node (in (24b), as well as in other figures below, we do not repeat an element when it occurs more then once in the same node of one and the same segment, for we assume that adding a specific element to a node that already contains that element has no phonological nor phonetic consequence).

(24) *s-ĕj* → [sɛj]



In contrast, /r/ has the element A in the manner node, and nothing else; this is shown in (25a).[18] When concatenated with *-ĕj*.CMPR, the palatalizers associate to the root node containing /r/, resulting in a structure that features the elements H and A in the manner node, and the element I in the place node (25b).

(25) *r-ĕj* → [r̝ɛj]



Thus, the geometrical difference between the palatalized alternants of /s/ and that of /r/ account for the different phonetic interpretation of an apparently similar set of elements, which surface as [s] or as [r̝], respectively.[19]

4.3.3 Labial consonants

As we observed above, the Czech labials seem to be unable to palatalise: either the palatal element is pronounced as a separate glide, or it remains unpronounced.

[18]Alternatively, assuming that the place node depends on the manner node (Botma 2004), one could argue that the element A percolates from the place to the manner node, and is thus found in both nodes.

[19]Note that [r̝] is commonly classified as a fricative consonant, which supports the hypothesis that the palatalized alternants of /r/ contain the element H in their manner node.

In addition, the labials seemed to necessitate that we distinguish not three but four patterns of palatalization. Yet we have claimed that there are only three different palatalizers, as shown in Table 16 (repeated from Sect. 4.1).

Table 16: Labial plain and palatalized consonants (input)

plain		PAL$_1$ adds I		PAL$_2$ adds H.I				PAL$_3$ adds H.I̲	
				CMPR *-ěj*		CAUS *-i*			
/p/	ʔ.U	/pi/	ʔ.U–I	/pi/	ʔ.U–H.I	/p/	ʔ.U–(H.I)	/p/	ʔ.U–(H.I̲)
/f/	H.U	/fi/	H.U–I	/fi/	H.U–H.I	/f/	H.U–(H.I)	/f/	H.U–(H.I̲)
/m/	L.U	/mɲ/	L.U-L.I	/mɲ/	L.U-H.I	/m/	L.U–(H.I)	/m/	L.U–(H.I̲)

In this section, we shall explain how the underlying representations of the table lead to the observed output. What we need to derive is (i) the failure of the palatalizers to modify a base-final labial consonant, (ii) their realisation as a "fully developed palatal element" instead, (iii) the (apparent) failure of the palatalizers to surface at all with certain suffixes, and (iv) the appearance of /ɲ following /m/.

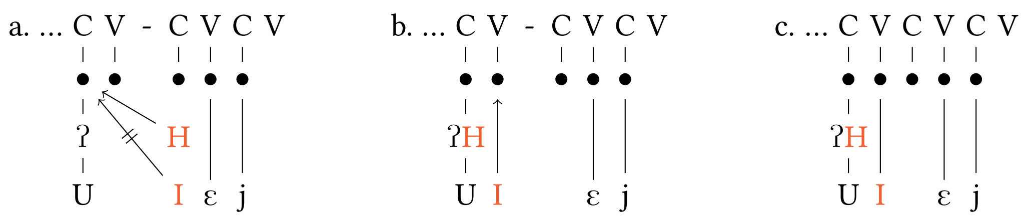
Let us start from the derivation of forms in which the palatalizer surfaces as a "fully developed palatal element" following an obstruent (represented as /pi/ and /fi/ in the PAL$_1$ and PAL$_2$ columns of Table 16). The derivation is shown in (26), with a form ending in /p/ that concatenates with *-ěj*.CMPR.[20] The crucial difference between velar and coronal consonants on the one hand, and labial consonants on the other is that the latter contain the element U, which in Czech cannot merge or coalesce with I to create a complex place element IU (Janků 2022).

Thus, when PAL$_2$ concatenates to a base ending in /p/ and its floating elements look for the leftmost available root node, H associates to the root node

[20] As in the case of velar consonants, we only provide the derivation of forms ending with a voiceless stop. The derivation of its voiced counterpart and of the voiceless and voiced labial fricative is identical to the one of the voiceless stop, provided one adds L̲ for voiced counterparts, and replaces ʔ with H for the fricatives. We do not provide the derivation of velar-ending forms followed by a PAL$_1$ morpheme such as *-ě*.LOC/DAT.F.SG either, as it would be identical to that of forms followed by *-ěj*.CMPR.

hosting /p/, i.e. ʔ.U, but I does not, for it is repelled by the U element of /p/ (26a). As a consequence, I tries to associate to the second-leftmost slot, i.e. the base-final V. As nothing prevents this, the palatalizer associates with that slot (26b), surfacing as /i/ (26c).[21]

(26) *p-ěj* → [piɛj]



This analysis implies that the 'fully developed palatal element' following labials is a vowel, rather than a consonant (whence our notation *p-ěj* → [piej] in the caption of (26)). We therefore expect it to sound and behave as such, and there is indeed some empirical evidence that this is correct. In Czech, vowels in infinitives undergo templatic lengthening (Caha & Scheer 2008). Examples of this are given in Table 17a. However, the vowel *ě* never lengthens, as shown in Table 17b. M. Ziková & A. Poĺomská (p.c.) suggest that this is because *ě* is already long. This would be precisely what is implied by our analysis in (26),

[21] Note that in (26)a, Gussmann & Kaye's (1993) reduction does not apply. It thus looks like reduction applies after other operations (such as spreading) have applied, as a repair strategy that removes an ill-formed structure. The motivation that (Gussmann & Kaye 1993: fn. 8) present for the reduction operation suggests that it 'could be derived from the empty syllable constraint or from OCP'. In the structure in (26)a, this ill-formedness is bled by the association of the palatalizer to the otherwise empty base-final V slot. The hypothesis that reduction applies after other operations applied is not widely assumed. Whereas Gussmann & Kaye (1993) do not take any position on the matter, Ulfsbjorninn (2022, 2024) claims that reduction is the first operation applying upon concatenation. Ulfsbjorninn also claims that reduction is parametric, whereas Gussmann & Kaye (1993) claim that it is universal. This suggests that the nature of this operation is still unclear. Be that as it may, note that operation ordering presupposes a serial derivation. However, if we assume instead an OT-style parallel computation (which is not incompatible with strict CV, see e.g. Cavirani 2015, Faust & Scheer to appear and references therein), this problem dissolves. Furthermore, note that our assumption about there being reduction in forms such as those in Figures (18), (20), (24), and (25) is not strictly necessary, as the base-final empty V slot could be governed by the non-empty V slot of the suffix. If one assumed a serial derivation, this would imply that government applies after spreading, otherwise the palatalizer could not be associated with the base-final V slot in (26). This is not unheard of, and it is in fact tacitly assumed by influential analyses such as Larsen (1998) (see Cavirani 2022b: fn. 14 for comments).

where the melodic content of the grapheme *-ě* is associated with two V-slots. In contrast, if I were associated to the C slot, i.e. the root node of /p/ generating a complex segment /p^j/, the length property of *-ě* would be quite unexpected. The phenomenon of Output Invisibility following labials (see Table 16) would also be hard to explain.

Table 17: PST∼INF alternation

	PST	INF	gloss
a.	C**V**C-l	C**VV**C-t	
	n**e**s-l	n**é**s-t	'bear'
	p**i**-l	p**í**-t	'drink'
	kr**a**d-l	kr**á**s-t	'steal'
b.	C**ie**C-l	C**ie**C-t	
	sp**ě**-l	sp**ě**-t	'support'
	p**ě**-l	p**ě**-t	'sing'

A further piece of evidence suggesting that *ě* does not lengthen comes from Czech dialect data discussed in Caha et al. (2024). In the dialects of Central Morava, four patterns of comparative adverb formation are attested, as shown in Table 18. Classes I and II are characterised by the apparent absence of any overt adverbial ending, the morphemes *-ěj* and *-š* in the rightmost (adverbial) column also being found in the adjectival comparative (middle column). However, Caha et al. (2024) argue that there is an underlying adverbial morpheme nevertheless, which can be seen at work in the classes III and IV, where in the adverbs the final vowel of the root is lengthened. Moreover, if they end in a nonlabial consonant, this consonant is palatalised (*drah-ý* 'expensive' → *drážˇ*). This same palatalising and lengthening morpheme is present in classes I and II, but its effects are invisible, since the preceding consonant (*-š*) is already palatal, and the preceding vowel (in this case *-ě*) fails to lengthen. This is, then, another case where *-ě* does not lengthen in a context where other vowels do. From the present perspective, the reason for this is that *-ě* is already long.

Let us now show the derivation of the forms where the palatalizer surfaces as a palatal nasal consonant. As observed by Short (1993: 459), "after bilabial /m/ nasal resonance extends over both segments". In strict CV terms, this means that the L element encoding nasality spreads to the following C slot. The fact that the nasal consonant following /m/ is /ɲ/ suggests that what spreads is the

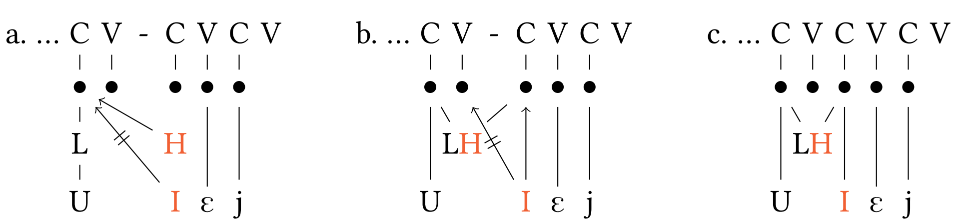
Table 18: Central Morava comparative adjectives and adverbs (Caha et al. 2024: 15)

	POS ADJ	CMPR ADJ	CMPR ADV	GLOSS
I.	chab-ý	chab-**ěj**-š-í	chab-**ěj**-š	'weak'
II.	hrub-ý	hrub- š-í	hrub-**ěj**-š	'rough'
III.	slab-ý	slab- š-í	sláb- š	'weak'
IV.	drah-ý	dra*ž*- š-í	dráž	'expensive'
	dobr-ý	lep- š-í	líp	'good'

L element only, rather than the root node hosting it.[22]

The example (27) shows the consecutive steps in the derivation. The floating I of the palatalizer cannot associate to the root node containing L and U because it is repelled by U; at the same time, the floating H of the palatalizer can associated leftward (27a). L and H join in the manner node, and this manner spreads over two consecutive C nodes (27b). As the representation shows, this spreading prevents I from associating to the root node of the final V slot of the base. The floating element I therefore associates to the next leftmost available root node, which is the one of the initial C of *-ěj*.CMPR, which already hosts LH. As a consequence, the palatalizer affects the nasal occurring in the suffix-initial C slot, which surfaces as [ɲ] (27c).

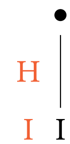
(27) m-ěj → [mɲɛj]



[22]We remain agnostic about the reason why L spreads, but we note that nasals, as well as other sonorants, seem to have a propensity for spreading and generate (fake) geminates. This is what is observed e.g. for French, where no other consonant than /m n l/ can be (optionally) pronounced as long (N. Faust p.c.). As we will see below (Sect. 4.3.4), this is indeed what we claim happens with /l/. Note that the spreading of L does not follow from an alleged impossibility for the palatalizer I to surface alone, as claimed for Serbian, where palatalizers surface as [ʎ] after (all) labials (Morén 2003), as in *tup* [tup] 'dull' → *tuplji* [tupʎi] 'duller'. In such cases, spreading is explicitly claimed to be triggered by the impossibility of the palatalizer to surface alone (Morén 2003: 66). This is clearly not the case in Czech, as the palatalizer can surface alone, witness (26).

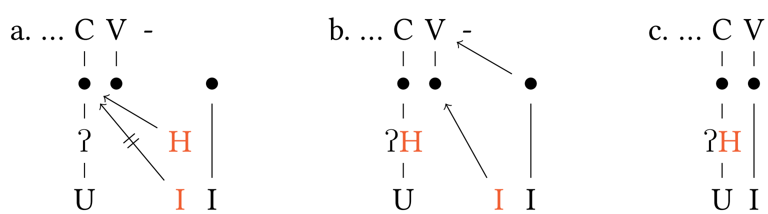
We now need to show why, in some cases, the palatalizer fails to surface. As we discussed above, this happens with the morpheme expressing causativity, repeated in (28).

(28) -*i*.CAUS



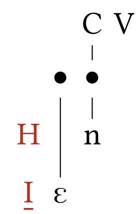
As shown, this morpheme consists of two floating elements, i.e. the palatalizer PAL_2 H and I, and the floating vowel /i/, which consists of a root node linked to the element I. As in the examples discussed above, when concatenated with a base, the floating elements scan the resulting phonological string left to right looking for an available root node. If the latter contains a labial, its U element blocks the association of I, while no issue arises with H, which associates to the root node and joins ʔ (29a). The palatalizer then turns to the next available slot i.e. the root node of the base-final V, and since there nothing to prevent its association, it links to that. The same holds for the floating segment /i/, which joins the palatalizer in the base-final V (29b). As the floating /i/ and the palatalizer I are identical, their merger generates an identical segment, which surfaces as [i] (29c).

(29) p-i → [pi]



The same effect can be observed when a base ending in labial is concatenated with a PAL_3 such as -*ěn*.PASS.PTCP, repeated in (30).

(30) *-ěn*.PASS.PTCP - PAL_3

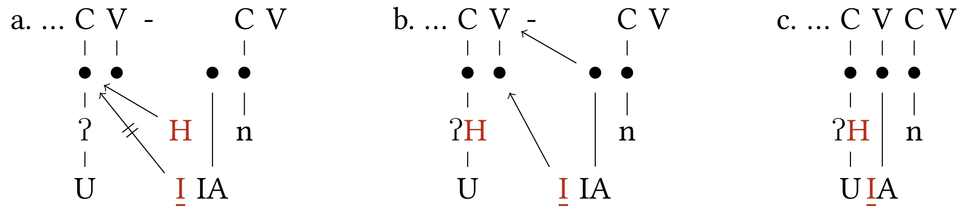


In this case as well, the H of PAL_3 associates with the root node containing ʔ.U, while I̲ cannot do that because of the presence of U (31a). I̲ therefore associates to the root node of the base-final V slot, where it is joined by the floating root node of the /ε/ vowel of the suffix, shown as IA in (31b). This results in the set of elements I, I̲, and A, which surfaces as [ε] (31c); as in (29c), we collapse the two I's.[23,24]

[23]Czech has a 5-vowel system, and shows no mid-high ∼ mid-low contrast. This suggests that headedness does not need to be recruited to account for a contrast between two different degrees of mid vowels (e.g. /ε/ I̲A *vs* /ε/ IA̲). As a consequence, we maintain that the merger of A with I or I̲ both result in the only front mid vowel of Czech, i.e. /ε/.

[24]As noted by an anonymous reviewer, this would imply that if a palatalizing suffix starts with /a/, the latter should surface as [e] due to the merger of I and A. As we reported in fn. 5, this is not the case. When suffixes starting with a back vowel follow a stem ending in labial, the vowel of the suffix does not change, nor does the palatalizer surface independently as a glide. A way to derive such pattern would be to adopt the vowel-before-vowel deletion approach proposed for Russian by Jakobson (1948) (see also Matushansky 2024 and Cavirani & Matushansky in preparation). In such a scenario, the two 'vowels' would be the floating palatalizer on the one hand and the floating back vowel of the suffix on the other, which would both compete for the same V slot. The latter would then outcompete the patalizer, which would lead to the palatalizer's Output Invisibility. This approach could also extend to the other two cases discussed in this section, i.e. *-i*.CAUS and *-ěn*.PASS.PTCP. In the alternative analysis we have just sketched, rather than merging with the floating vocalic segment *-i* and *-ě*, the palatalizers would just fail to associate to the stem-final V slot because outcompeted by *-i* and *-ě*. Where the palatalizer does surface as a glide, as it does before comparative *-ěj*, it would do so because the suffix provides an empty C slot for the glide to associate with. We shall not explore this alternative approach to cases of Output Invisibility any further here.

(31) p-ěn → [pɛn]



In Table 19 we present an overview of the labials, but different from Table 16, this Table shows the output situation of the derivations, such as we have presented them in this section.

Table 19: Labial plain and palatalized consonants (output)

plain		PAL$_1$ adds I		PAL$_2$ adds H.I				PAL$_3$ adds H.I̱	
				CMPR *-ěj*		CAUS *-i*			
/p/	ʔ.U	/pj/	ʔ.U–I	/pj/	ʔH.U–I	/p/	ʔH.U	/p/	ʔH.U
/f/	H.U	/fj/	H.U–I	/fj/	H.U–I	/f/	H.U	/f/	H.U
/m/	L.U	/mɲ/	L.U-L.I	/mɲ/	LH.U-LH.I	/m/	LH.U	/m/	LH.U

4.3.4 The lateral consonant

In this section, we discuss the consonant that, despite being commonly characterized as coronal, behaves like a labial: /l/. As a coronal, it could be expected to have the element A and/or I. The diachrony in many cases supports that, like Latin /l/ becoming /i/ in Italian (e.g. Lat. ***plenum*** 'full', ***flora*** 'flower', ***oculus*** 'eye' → It. *pieno*, *fiore*, and *occhio*). However, there is also evidence that /l/ (like labials) contains U. For example, in the Italian dialect of Pontremolese, Latin /l/ in codas developed into either /r/ (i.e. A or I) or /w/ (i.e. U) depending on the context: when occurring between segments containing A, /l/ developed into [w], e.g. Lat. *ca**l**(i)du(m)* 'warm.M.SG' → [kawd], whereas when the surrounding segments contain U, /l/ developed into [r], e.g. Lat. *co**l**(a)phu(m)* 'strike' → [kurp] (Cavirani 2015). This development of /l/ into an *u*-like sound is widely documented cross-linguistically, and suggests that laterals contain U. This is confirmed by some diachronic evidence from within Slavic, where /l/ developed into an [u]-like sound in Polish, e.g. *g*[u]*ova* 'head', from Proto-Slavic **glava*. The difference between clear ([i]-like) and dark ([u]-like) /l/ may also be thought of in terms of the presence of U for dark /l/. More precisely, in languages that display such dark variants, they are represented as containing A and U (Backley 2011).

The Czech lateral /l/ is described in traditional grammars as a coronal, more specifically an apico-alveolar lateral approximant of light or neutral quality (Hála 1962, Dankovičová 1999). However, as claimed by Šimáčková (2009: 125), this pronunciation of /l/ is the one assumed by "language professionals who work with *orthoepic* rules". This pronunciation is frequently replaced by a darker pronunciation, especially in weak contexts, such as coda position, and speakers of younger generations, which are apparently more liberal with respect to prescriptive rules.

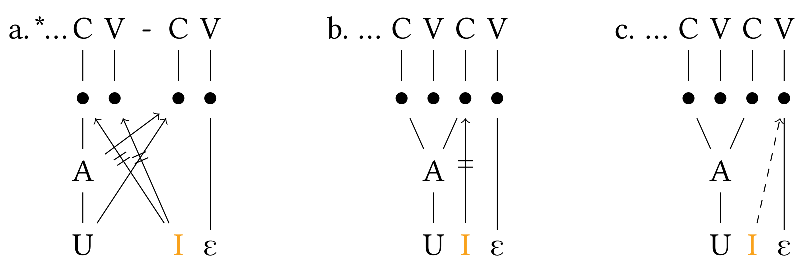
This description suggests two possibilities for the analysis of Czech /l/, one where /l/ does not have U, and one where it does. If we take it to contain I and no U (as suggested by a reviewer), then the analysis of the absence of palatalization with /l/ can be kept relatively simple, as shown in Table 20a. The small palatalisation will have no effect, since it does not change the underlying representation. That leaves us with three different underlying representations. This analysis requires us to assume that these all map onto the same phonetic interpretation. Under this approach, there is no principled connection between the behaviour of lateral /l/ and the labials.

Table 20: Lateral plain and palatalized consonant

	plain		PAL_1 adds I		PAL_2 adds H.I		PAL_3 adds H.$\underline{I}$	
a.	/l/	A.I	/l/	A.I	/l/	AH.I	/l/	AH.$\underline{I}$
b.	/l/	A.U	/l/	A.U–I	/l/	AH.U–I	/l/	AH.U–$\underline{I}$

What the analysis does not explain, however, is why we get the dark, [w]-like pronunciation of /l/ in weak contexts reported by Šimáčková (2009). We therefore adopt the alternative analysis where Czech lateral /l/ does contain U, and where there is consequently a representational commonality with the labials. For concreteness, we shall assume it is represented as A.U, where A represents the liquid class, and U the place. This analysis is represented in Table 20b. Under this approach, we expect /l/ to behave like other segments containing U, thus not to change its place and manner of articulation. This is indeed what we observe, for /l/ never turns into [ʎ] or [lj]. Formally, as we saw for the labial consonants in the previous section, the surfacing of a final /l/ followed by a palatalizer as [ʎ] is prevented by the impossibility of combining two segments containing U and I. This is shown in (32), where a stem ending in *-l* is followed by *-ě*.LOC/DAT.F.SG. Once the attempt of I to associate to the root node containing U has failed, in principle, there could still be the possibility for the I to associate to the base-final V slot, generating [lj], rather than the attested [l]. As shown in (32a), we hypothesize that laterals behave like labial nasals, which spread to the following C slot (see footnote 21). Because of this association, the palatalizer cannot associate to the root node of base-final V slot, as that would require crossing an already established association line. The other possibility the palatalizer has to surface is associating to the root node of the suffix-initial C slot. However, if this root node is associated to the lateral, the palatalizer cannot associate there either, as the lateral contains U, which repels the palatalizer's I (32b). As a consequence, the palatalizer stays afloat (or perhaps merges with the root node containing /ɛ/, i.e. IA, yielding IA, see (31c), as shown in (32c).[25]

(32) l-ě → [lːɛ] → [lɛ]



Note that the representation in (32) would normally predict that the lateral should

[25]Note that the derivation in (32), as well as the one in (27), does not necessarily presuppose a serial derivation in which the spreading of A.U to the C slot of the suffix precedes, and therefore bleeds, the association of the palatalizer to the stem-final V node. In fact, the desired output can be obtained within an OT-like parallel evaluation, where the successful candidate in (32)b is selected because it incurs the lower number of violations of high-ranked constraints. More precisely, assuming a constraint triggering the realization of the palatalizer — e.g. Realize Morpheme(I) — and another one forcing AU spreading — e.g. Spread(AU) (McCarthy 2011) —, we would need to assume that Spread(AU) is ranked higher than Realize Morpheme(I). The two representations conflated in (32)a would violate constraints ranked higher than Realize Morpheme(I), e.g. a markedness constraint penalizing IU — e.g. *IU — and the No-Crossing Constraint (NCC; Goldsmith 1976).

surface as long, for it is associated to two C slots. We argue that the pronunciation of this structure as phonetically long is prevented by a general ban Czech has on geminates (Bičan 2011), which forces /l-l/ to surface as [l] (this can be understood as a case of virtual geminate; Sègèral & Scheer 2001).

5 Conclusion

In this paper we examined different palatalization patterns in Czech. Following Scheer (2001), we identified two patterns: 'small' (e.g. /k/ → [ʦ], /t/ → [c]) and 'big' palatalization (e.g. /k/ → [ʧ], /t/ → [ʦ]). In line with what observed in typological literature (Bateman 2007, Kochetov 2011), Czech labials are not modified. Instead, palatalization may manifest itself as the surfacing of a front vocoid after the labial, which, if the base is nasal, gets nasalised ([ɲ]). Alternatively, labials may show Output Invisibility, i.e. the absence of any visible sign of palatalization.

We extended Scheer's (2001) typology and identified a third pattern, which we dubbed 'medium' palatalization. The difference between medium palatalization and the other two patterns is not so much in terms of how much the base segment is modified, but rather in terms of the classes of base segment affected by palatalization: whereas velar bases surface as if they were affected by big palatalization, coronals show the effect of small palatalization. Furthermore, the medium palatalization can be further split into two patterns, depending on whether labials pattern with small (i.e. presence of [i] or [ɲ]) or big (i.e. absence of [i] or [ɲ]) palatalization (Tables 3 and 4).

Besides this empirical contribution, we have provided a formal modeling of the patterns that exploits the representational technology of strict CV (Lowenstamm 1996, Scheer 2004) and Element Theory (Kaye et al. 1985, Harris 1994, Harris & Lindsey 1995, Backley 2011, Scheer & Kula 2017), arguing that the three degrees of palatalization are triggered by three different palatalizers. We represented those as floating elements contained in the suffixes that concatenate with the base. These three sets of floating elements stand in an inclusion relation, i.e., from small (I) over medium (H.I) to big (H.I̲). The hypothesis that these palatalizers are floating elements, and that suffixes containing them might also contain floating segments, i.e as sets of elements that behave like units, required us to find a representational way to distinguish between floating elements and floating segments. We departed from standard strict CV and resorted to the root node, which mediates between the element level of representation and the skeletal level, i.e. the one hosting the C/V slots (for a similar proposal, see John 2014, Ziková 2018, Scheer 2022, Cavirani 2022b, Faust to appear).

The hypothesis that the palatalization patterns are triggered by floating palatalizers, rather than by the vowel of the suffix, allowed us to account for the fact that (i) palatalization can be triggered by back vowels too (Table 8), and (ii) it is not triggered by all front vowels (Table 9), and (iii) similar-sounding vowels trigger different palatalization patterns. This is in line with other accounts of palatalization in Slavic (see e.g. Ketner 2005 for Czech, Gussmann 1992, Zdziebko 2015 for Polish, and Bidwell 1962, Iosad & Morén-Duolljá 2010, Padgett 2010 for Russian), and provides support for the hypothesis that "featural representations [...] possibly reflect local generalizations, specific to certain morphological domains or lexical strata (as, for example, in cases of multiple palatalization processes targeting the same consonants)" (Kochetov 2011).

As noted by Kochetov (2011), throughout its long history, the formal modeling of palatalization has been affected by "difficulties [which] stem from a [...] fundamental problem – the persistent use of traditional [feature-geometrical] representations (with some modifications), which were assumed to be inviolable, universal, and innate". We maintain that our element-theoretic analysis improves on previous accounts, and provides a neat formal modeling of the palatalization patterns observed. This is obtained by exploiting the substance-free take on phonology (Reiss 2017, Reiss & Volenec 2022, Scheer 2019, Chabot 2022, 2023) and, to a lesser extent, on Element Theory. This allows us to devise a representational account of the three palatalization patterns of Czech, as well as of its consonant inventory, where, despite the phonetic appearances (Table 21), the palatalized consonants that pattern together all contain the same palatalizers (Table 22).

Table 21: The patterns of small, medium, and big palatalization: phonetics

	small	medium	big
/k/	/ts/	/tʃ/	/tʃ/
/x/	/ʃ/	/ʃ/	/ʃ/
/t/	/c/	/c/	/ts/
/s/	/s/	/s/	/ʃ/
/n/	/ɲ/	/ɲ/	/ɲ/
/r/	/r̝/	/r̝/	/r̝/

Table 22: The patterns of small, medium, and big palatalization: phonology

	PAL$_1$	PAL$_2$	PAL$_3$
ʔ H	ʔ.I H.I	ʔH.I H.I	ʔH.$\underline{\text{I}}$ H.$\underline{\text{I}}$
ʔ.A H.A	ʔ.AI H.AI	ʔH.AI H.AI	ʔH.A$\underline{\text{I}}$ H.A$\underline{\text{I}}$
L A	L.I AI	LH.I AH.I	LH.$\underline{\text{I}}$ AH.$\underline{\text{I}}$

We also showed that Element Theory and strict CV allows for a neat modeling of (i) the unexpected behavior of labials (Table 12), whose U element clashes with the I element of the palatalizers, thereby preventing the merger of the latter with base-final labials, (ii) the surfacing of the palatalizers as self-standing segments, or their complete absence, which is related to the CV profile of the suffix, and (iii) the fact that, when the palatalizer does surface, it can do so either as a vowel [i], or as a palatalized nasal. The appearance of the nasal is due to the spreading of the nasal element L from root node in the base-final C slot to the one of the suffix-initial C slot, where it joins the palatalizer (Section 4.3.3).

Finally, the hypothesis that the Czech lateral /l/ contains the U element allowed us to account for the fact that it behaves like a labial consonant, in that it resists palatalization (Sect. 4.3.4).

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